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Cooperative control of a tethered drone and surface vessel for sustained presence at sea

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Contents

Acronyms . . . vii

1 Introduction **1**

1.1 Motivation . . . 1
1.2 Problem Statement . . . 2
1.3 System Overview . . . 3
1.4 Expected Contributions . . . 4
1.5 Thesis Structure . . . 4

2 State of the Art and Related Work **5**

2.1 Modeling and System Identification . . . 5
2.1.1 White-Box, Grey-Box, and Black-Box Modeling . . . 5
2.1.2 Tether Dynamics Modeling . . . 6
2.1.3 Optimization-Based Estimation . . . 6
2.1.4 Related Work on Tethered Vehicle Modeling . . . 7
2.2 Control Strategies for Autonomous Vehicles . . . 7
2.2.1 Classical and Modern Control . . . 7
2.2.2 Predictive and Optimization-Based Control . . . 8
2.2.3 Active Winch and Tension Control . . . 8
2.2.4 Related Work on Tethered Vehicle Control . . . 8
2.3 Cooperative Coordination and Systems . . . 9
2.3.1 Centralized vs. Distributed Coordination . . . 9
2.3.2 UAV-USV Cooperative Systems . . . 9
2.3.3 Related Work on Maritime Cooperative Robotics . . . 10
2.4 Path and Trajectory Planning . . . 10
2.4.1 Offline and Online Planning . . . 10
2.4.2 Related Work on Trajectory Planning . . . 11

3 Proposed Methodology **13**

3.1 System Modeling . . . 13
3.1.1 Reference Coordinate Frames and Notation . . . 13
3.1.2 WAM-V Surface Vessel Model . . . 14

3.1.3 x500 Quadrotor Model . . . 18
3.2 System Control . . . 23
3.2.1 Cooperative Control Design . . . 23
3.2.2 Nonlinear Model Predictive Control . . . 25
3.2.3 USV NMPC . . . 27
3.2.4 UAV NMPC . . . 31

4 Implementation **37**
4.1 Simulation Environment . . . 37
4.1.1 World . . . 38
4.1.2 Wave Adaptive Modular Vessel . . . 38
4.1.3 X500 Quadrotor . . . 40
4.1.4 Tether . . . 42
4.1.5 Tether Winch . . . 44
4.2 Model Validation . . . 46
4.2.1 USV Dynamics Validation . . . 46
4.2.2 UAV Dynamics Validation . . . 48
4.2.3 Tether Model Validation . . . 52
4.3 Controller Implementation . . . 55
4.3.1 Embedded Optimization Framework (CasADi & Acados Pipeline) . . . 55
4.3.2 USV Controller Implementation . . . 55
4.3.3 UAV Controller Implementation . . . 55
4.4 Maneuvers and Missions . . . 55
4.4.1 Coordinated Maneuver Primitives . . . 55
4.4.2 Autonomous Missions and Coordination Logic . . . 55
4.5 Software Architecture and Dashboard . . . 55
4.5.1 Operator Web Dashboard and Mission Management . . . 55
4.5.2 Integrated Software Architecture and System Dataflow . . . 55

5 Conclusions **57**
5.1 Summary of Achievements . . . 57
5.2 Future Work . . . 58

Bibliography **59**

Acronyms

DOF	degree of freedom
EMSA	European Maritime Safety Agency
LTI	linear time-invariant
MPC	Model Predictive Control
NMPC	Non-linear Model Predictive Control
PID	proportional–integral–derivative
RMSE	root-mean-square error
ROS	Robot Operating System
UAV	Unmanned Aerial Vehicle
USV	Unmanned Surface Vessel
VRX	Virtual RobotX
WAM-V	Wave Adaptive Modular Vessel

Chapter 1

Introduction

This chapter introduces the thesis, presenting the context, motivation, and objectives of the research, followed by an overview of the system architecture, the expected scientific and engineering contributions, and the overall structure of the dissertation.

1.1 Motivation

Saving lives at sea, protecting marine ecosystems, and securing maritime infrastructure all rely on one essential capability: understanding what is happening, where, and when. In vast and dynamic maritime environments, this requires maintaining reliable, high-resolution situational awareness over long periods. The European Maritime Safety Agency (EMSA) recognises this challenge and highlights the need for continuous monitoring and locally persistent operations to enable effective maritime surveillance and response [1].

Different maritime assets meet this challenge in complementary ways. Surface vessels give a physical presence at sea, with long endurance and large payload capacity [2, 3]. However, their range is limited by line of sight, and operations are expensive [4]. Aerial platforms, such as manned aircraft and Unmanned Aerial Vehicles (UAVs), allow fast deployment and elevated sensing [5]. However, aircraft are costly, and UAVs suffer from low endurance and unreliable communication [6].

Together, these assets form a connected system rather than separate solutions. Each one compensates for what the other lacks. Unmanned Surface Vessels (USVs) provide long-term presence at sea at much lower cost than crewed vessels, allowing wide and affordable patrols. UAVs extend this reach by sensing beyond the horizon from the air. Working as a team, they maintain continuous and local maritime awareness. This motivates the study of coordinated USV–UAV systems, where cooperation creates mission capabilities that neither platform can achieve alone in complex and changing environments.

A natural way to strengthen this cooperation is to add a physical tether between both platforms. The tether provides continuous power and reliable communication between the USV and UAV [7]. By placing the energy supply and communication systems on the USV, it overcomes two main limits of maritime

UAVs: short endurance and weak long-range communication. This approach has recently transitioned from academic concepts to real-world industrial systems; for instance, the marine robotics company Exail partnered with Elistair to integrate a tethered quadrotor directly onto the DriX USV catamaran for persistent maritime surveillance [8]. Similarly, ACUA Ocean and Teledyne FLIR Defense demonstrated an integrated tethered aerial-surface system for the UK Ministry of Defence to support long-endurance maritime security and target detection [9].

Figure 1.1 illustrates the configuration of the tethered USV–UAV system considered in this work.

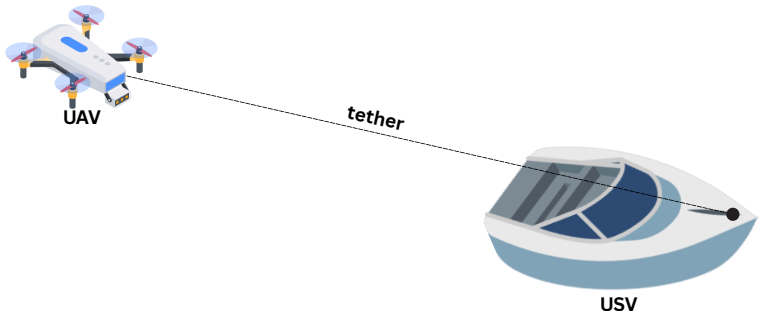


Figure 1.1: Schematic of a tethered USV–UAV system.

The main challenge of this work is to design a control framework that enables two autonomous vehicles, connected by a tether, to operate as a single coordinated system. The controller must manage not only the motion of both vehicles but also account for the tether constraints and dynamics, ensuring safe operation and robustness to wind, waves, and obstacles.

The USV and UAV operate under very different dynamics. The USV moves slowly, dominated by hydrodynamic drag and inertia, while the UAV reacts quickly, with motion evolving on much shorter time scales. Coordinated control is challenging because the controller must handle these different response rates and actuation dynamics consistently.

The maritime environment adds further complexity. Wind, waves, and water motion constantly disturb both platforms, making robustness to disturbances essential. Nearby structures, vessels, or obstacles can restrict their movement and risk tether entanglement. Effective cooperation relies on continuous information exchange and synchronised decision-making between both agents.

Most existing tethered UAV systems focus on power delivery and communication, with operation often remaining manual or semi-autonomous. In research prototypes, the aerial and surface vehicles are usually decoupled, the tether is simplified or linearised, or the surface platform is treated as fixed. These assumptions make control design easier but reduce realism. Few cooperative control approaches explicitly model the tether while coordinating both agents under realistic maritime conditions.

1.2 Problem Statement

This work addresses the problem of controlling a cooperative system composed of an autonomous surface vessel (the WAM-V USV) and a physically connected quadrotor (the x500 UAV) operating in a

coordinated manner. The objective of such a system is to autonomously execute maritime missions by following predefined waypoints or continuous reference trajectories, while respecting the physical and operational constraints of both platforms and of the tether that connects them.

To achieve this objective, the control framework must be capable of handling different phases of operation and mission requirements. This includes the definition of multiple control modes that allow the system to perform manoeuvres such as takeoff and landing, as well as different forms of coordinated motion during navigation. Both NMPCs actively prevent the tether from exceeding the maximum admissible separation distance, utilizing soft constraints and slacks, while exchanging predicted trajectories to coordinate the motion of the aerial vehicle and the dynamic vessel deck.

1.3 System Overview

Interaction with the system is defined at the mission level. The operator selects a mission, such as patrolling a given area, following a moving target, or visiting a set of waypoints that represent regions of interest. This mission information is received by a behaviour management layer, which selects the active control mode and handles mission logic, including switching between different relative configurations of the UAV with respect to the USV, as well as executing specific manoeuvres such as takeoff and landing.

Based on the selected mission and active control mode, the system generates coordinated motion for both the USV and the UAV. This process combines planning and control to compute trajectories and actuation commands that satisfy vehicle and tether constraints while pursuing the mission objectives. The WAM-V USV acts as the leader planar platform carrying an active winch system on its deck. The winch dynamically spools the tether in or out to manage slack and cable tension. The x500 UAV follows a relative orbit or observation trajectory above the USV, using its elevated perspective to record visual data.

Figure 1.2 presents a high-level block diagram of the proposed system, showing the interaction between the coordinator, the decentralized NMPC solvers, the vehicles, and the co-simulation interfaces.

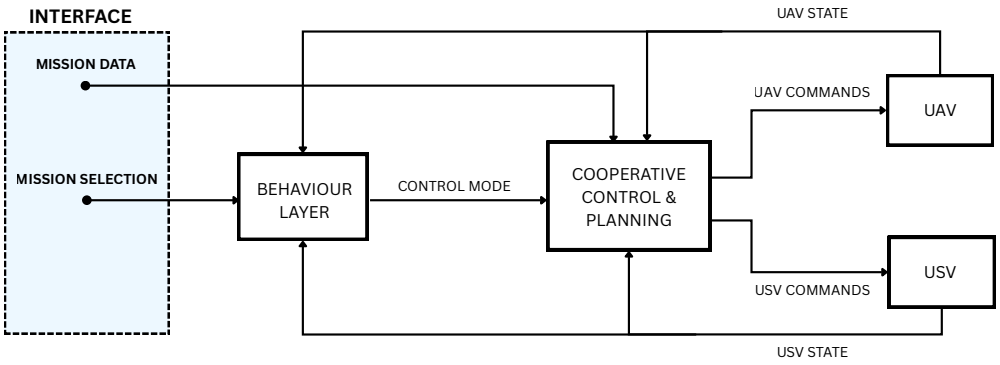


Figure 1.2: High-level block diagram of the proposed cooperative control system.

1.4 Expected Contributions

This thesis contributes to the design, implementation, and evaluation of cooperative control solutions for tethered UAV–USV systems, to enable autonomous and persistent maritime missions. The main contributions of this work are summarised as follows:

- A unified, C++ ROS 2 control pipeline utilizing Acados and HPIPM to generate and execute real-time Nonlinear Model Predictive Control solvers for both surface tracking and aerial guidance.
- The modeling and experimental identification of the UAV inner-loop attitude response lag ($\tau = 0.15\,\text{s}$) using PX4 doublets flight data.
- The design and implementation of a virtual winch controller operating in both geometric and tension-feedback PD modes to prevent cable slackness and tension spikes.
- A realistic co-simulation setup in Gazebo Harmonic, integrated with the PX4 Software-in-the-Loop (SITL) firmware and the MoorDyn physics engine for multi-body lumped-mass line representation.
- The verification of NMPC constraint enforcement (states, inputs, and tether separation limits) and evaluation of solver latency.
- The validation of the entire cooperative control stack under an integrated search and pursuit mission targeting a vessel teleoperated in real-time.

1.5 Thesis Structure

The remainder of this dissertation is organized as follows:

- **Chapter 2 (Literature Review and State of the Art)** reviews existing research in cooperative marine-aerial robotics, tethered drones, and predictive control algorithms.
- **Chapter 3 (System Modeling)** derives the kinematic and dynamic models of both vehicles, as well as the catenary tether force model.
- **Chapter 4 (Cooperative Control Design)** details the decentralized NMPC mathematical formulation, cost functions, and physical constraints for both platforms.
- **Chapter 5 (Simulation Environment and Software Implementation)** introduces the physical characteristics of the virtual WAM-V and x500 platforms, the active winch spooling controller, the MoorDyn tether integration, the simulation setup, and the software modularity.
- **Chapter 6 (Results and Discussion)** evaluates the model validation, parameter identification, tracking precision, constraint enforcement, solver latency, and the target pursuit mission.
- **Chapter 7 (Conclusions and Future Work)** summarizes the findings of the thesis and outlines directions for future research.

Chapter 2

State of the Art and Related Work

This chapter is structured into four main sections, each representing a key element of cooperative system control. Each section presents the main techniques commonly used in control theory, together with subsections reviewing related work on topics closely related to this research.

The first section addresses *modelling and identification*, which provides the mathematical and experimental tools required to describe system dynamics and estimate relevant parameters, including tether and cable physics. The second focuses on *control strategies*, ensuring stability, accurate tracking, and robustness to disturbances at the individual agent level, including predictive control and active winch architectures. The third section concerns *cooperative coordination*, which enables communication and shared decision-making so that multiple agents can operate as a unified system. Finally, *path and trajectory planning* deals with the generation of feasible and safe motion plans that connect mission objectives with low-level control actions.

2.1 Modeling and System Identification

Understanding how a system evolves is a prerequisite for any control design. Modelling provides this understanding by offering a mathematical description of the system dynamics, capturing the dominant behaviours while deliberately neglecting secondary effects that are not essential for analysis or control purposes [10]. The level of detail of such a model is inherently linked to the available physical insight, the amount and quality of experimental data, and the specific objectives for which the model is intended.

2.1.1 White-Box, Grey-Box, and Black-Box Modeling

Depending on the available physical knowledge and data, modeling methodologies are classified into three main groups:

- **White-Box Modelling:** Based on first principles and physical laws that govern system behaviour. By expressing the dynamics through analytical formulations, often derived from Newtonian mechanics, hydrodynamics, or aerodynamics, this approach provides clear insight into stability, con-

trollability, and energy transfer. To preserve analytical tractability, simplifying assumptions such as linearisation or the neglect of higher-order effects are commonly introduced.

- **Grey-Box Modelling:** Lies between purely analytical and purely data-driven approaches. The structure of the model is defined using physical insight, while parameters that are difficult to determine analytically, such as hydrodynamic or aerodynamic coefficients, are estimated from experimental data. This combination preserves a clear physical interpretation while improving accuracy through calibration.
- **Black-Box Modelling:** Relies entirely on measured input–output data to describe system behaviour. Rather than using explicit physical laws, this approach identifies mathematical relationships that best fit the observed data. Typical examples include linear parametric models, state-space representations obtained from data, and learning-based methods.

2.1.2 Tether Dynamics Modeling

Modeling flexible tethers is critical since they introduce highly nonlinear forces and kinematic constraints on the vehicles:

- **Quasistatic Catenary Models:** Describe the steady-state profile of a heavy cable suspended between two points. These models use catenary equations to resolve cable sag and tension analytically, assuming the line is in static equilibrium at each step. While computationally efficient, they neglect dynamic effects such as transverse wave propagation and drag forces, making them unsuitable for rapid aerial manoeuvres.
- **Lumped-Mass Models:** Discretize the tether into a finite chain of nodes, or "lumped masses", connected by spring-damper segments. Each segment models axial elasticity, damping, and hydrodynamic drag, while the nodes resolve gravitational and buoyancy forces. This formulation, implemented in solvers such as MoorDyn, offers a high-fidelity representation of transient tether whip, slackness, and hydrodynamic drag under dynamic boundary conditions.
- **Finite-Segment and Multi-Body Models:** Formulate the line as a chain of rigid cylinders connected by spherical joints. This approach computes bending and torsional moments accurately, but requires solving high-index differential-algebraic equations, which increases the computational burden.

2.1.3 Optimization-Based Estimation

System identification offers an alternative path by constructing models directly from measured input–output data when physical knowledge is incomplete. Optimisation-based estimation focuses on identifying model parameters offline by minimising the mismatch between measured and predicted system outputs. This is typically achieved through cost functions based on least-squares formulations, including weighted least-squares and prediction-error methods (PEM) [11]. For nonlinear or constrained problems,

advanced optimisation schemes such as trust-region methods and constrained least-squares are often employed, allowing grey-box models to be calibrated while enforcing physical bounds and regularisation.

2.1.4 Related Work on Tethered Vehicle Modeling

In systems connected by a tether, the model of the overall system must account at least for the tether geometry to constrain the relative motion of the agents and avoid physically infeasible configurations that could lead to tether failure.

In [12], a buoy–UAV system is presented in which the tether is modelled explicitly as a geometric and dynamic constraint between the buoy and the aerial vehicle. The tether tension is computed from the coupled dynamics and incorporated as an external force acting along the tether direction in the UAV model. This formulation relies on simplifying assumptions, namely that the tether is inextensible, massless, free of hydrodynamic effects, and remains taut during coupled operation.

An alternative modelling perspective is proposed in [13], where tether-induced forces are not included explicitly in the system dynamics. Instead, their effect is lumped into an external disturbance acting on the UAV. This disturbance is then handled at the control level through active disturbance rejection, avoiding the need for explicit tether modelling while still compensating for its influence in real time.

In [14], a cooperative UAV–USV system is modelled for the purpose of manipulating a floating object. Both vehicles are connected to the object through individual tethers, and the full system is described using simplified white-box models. The formulation assumes inextensible and massless tethers that remain permanently taut.

In [15], a tether is not included, and both the UAV and the USV are described using simplified white-box models. The USV is represented by a planar marine vehicle model commonly used for path-following applications, while the UAV is modelled independently using a reduced-order dynamic model.

2.2 Control Strategies for Autonomous Vehicles

Control theory provides the mathematical framework for designing systems that regulate their behaviour through feedback, ensuring stability, accuracy, and robustness in the presence of uncertainty and disturbances.

2.2.1 Classical and Modern Control

Classical control is based on the representation of systems through transfer functions in the Laplace or frequency domain, under the assumption of linear time-invariant (linear time-invariant (LTI)) dynamics. Within this framework, feedback laws are typically implemented through proportional–integral–derivative (proportional–integral–derivative (PID)) controllers and frequency-domain compensators designed to shape the closed-loop response.

Modern control theory generalises these principles to systems that are multivariable, time-varying, or subject to strong coupling between states and inputs. It is formulated in the *state-space* domain, where

the internal dynamics of a system are described by sets of first-order differential equations. This representation enables systematic analysis of controllability, observability, and stability. Within this framework, optimal control theory defines feedback laws that minimise a performance index, such as the linear quadratic regulator (LQR) or robust control methods such as H_∞ [16, 17].

2.2.2 Predictive and Optimization-Based Control

Predictive and optimisation-based control methods determine control actions by solving an optimisation problem that predicts the system behaviour over a finite horizon. At each sampling instant, the controller minimises a cost function subject to system dynamics and operational constraints, typically penalising deviations from reference trajectories and excessive control effort. Only the first control action is applied, and the process repeats at the next time step, forming the receding-horizon principle that defines Model Predictive Control (MPC) [18].

Extensions to nonlinear systems led to the development of Non-linear Model Predictive Control (NMPC), which replaces linear models with nonlinear formulations and uses numerical optimisation techniques to compute control inputs in real time [19, 20]. To enable real-time embedded execution on high-rate systems (e.g. UAV flight control), modern applications rely on symbolic framework code generators (such as CasADi) that export native C solvers. These solvers utilize highly efficient quadratic programming kernels such as HPIPM and execution frameworks like the Sequential Quadratic Programming Real-Time Iteration (SQP-RTI) scheme, which performs a single optimization iteration per time step to minimize solver latency.

2.2.3 Active Winch and Tension Control

In tethered UAV operations, winch design is a core element:

- **Passive Tension Tethers:** Rely on spring-loaded spools or constant-torque slip rings. While simple and requiring no electrical control, they cannot adapt to fast vessel manoeuvres, leading to large tension spikes or excessive slack.
- **Active Spooling Winches:** Utilize motorized actuators regulated by microcontrollers. The winch dynamically adjusts the tether length based on geometric tracking (distance between platforms) or force-feedback algorithms.
- **Tension-Feedback and Active Heave Compensation (AHC):** Read force telemetry from a load cell. A closed-loop proportional-derivative (PD) controller regulates the winch spooling speed to maintain a target tension. In maritime environments, this functions as AHC, isolating the flying quadrotor from wave-induced vessel heave.

2.2.4 Related Work on Tethered Vehicle Control

In tethered UAV systems, the tether introduces additional forces and coupling effects that influence control design. In [12], this coupling is addressed through a nonlinear backstepping controller derived

directly from the white-box dynamics of the UAV–buoy system. The work in [13] adopts an alternative approach based on active disturbance rejection, where the tether influence is not explicitly modelled but treated as an external disturbance estimated in real time by a reduced-order observer.

In [14], cooperative manipulation with two tethered agents is controlled using a MPC formulation based on simplified white-box models of the UAV, the USV, and the floating object. A similar predictive strategy is presented in [21], where both the UAV and the USV are coordinated using NMPC, relying on known dynamic models.

In [15], the USV employs a line-of-sight guidance law combined with PID control for heading and speed regulation. The UAV motion is controlled using an NMPC formulation, which generates reference accelerations to track the desired relative trajectory. No joint or centralised control law is employed, and each vehicle is controlled using its own dedicated controller.

2.3 Cooperative Coordination and Systems

Cooperative coordination refers to the control and organisation of multiple agents working towards achieving goals through communication and joint decision-making.

2.3.1 Centralized vs. Distributed Coordination

Depending on how information is exchanged and the control authority is structured, cooperative architectures are broadly categorised as centralized or distributed:

- **Centralized Coordination:** A single decision node aggregates global information and computes control commands for all agents. This architecture facilitates global optimisation and ensures consistency of mission-level objectives, since the planner has access to full system states and constraints. However, centralized control scales poorly with team size and is vulnerable to communication delays or single-point failures.
- **Distributed Coordination:** Eliminates the need for a central planner by allowing each agent to make decisions based on locally available information and limited communication. Agents exchange state or intent data to achieve group-level objectives. Distributed coordination provides scalability and robustness to communication losses and agent failures, though it faces challenges related to synchronization and network stability.

2.3.2 UAV-USV Cooperative Systems

Cooperative deployments of UAVs and USVs combine the speed and altitude of aerial sensing with the weight-bearing capability and range of marine hulls:

- **Autonomous Landing on Moving Surfaces:** Focuses on landing a quadrotor on a moving deck (helideck) in waves. This requires precise relative state estimation (visual markers, RTK-GPS) and

predictive control to synchronize the UAV's landing velocity with the sea-state pitch/roll profile of the USV deck.

- **Mobile Recharging Stations:** The USV acts as a mother ship. Drones fly reconnaissance sweeps, return to land on the USV deck to swap or recharge batteries, and take off again, enabling long-term persistence at sea.
- **Cooperative Target Tracking:** Cooperative path-following algorithms steer the USV planar trajectory to intercept a target, while the UAV orbits above to provide continuous visual tracking and image feedback.

2.3.3 Related Work on Maritime Cooperative Robotics

In [21], cooperation is based on a tracking strategy in which the UAV is controlled to follow and land on a moving USV. Only the UAV is actively controlled, while the USV moves independently and provides its state as a reference. The interaction is therefore unidirectional, with the UAV adapting its motion to the USV.

In contrast, [14] proposes a centralised model predictive control approach in which the UAV and the USV are controlled jointly. A single optimisation problem computes the control inputs for both vehicles, explicitly accounting for the coupling introduced by the tether. This approach focuses on cooperative object manipulation rather than on mission-level coordination between the vehicles.

In [15], coordination is achieved through a leader–follower structure. The USV acts as the leader and follows a predefined path, while the UAV adjusts its motion to maintain a desired relative configuration with respect to the USV. Coordination is therefore based on reference sharing and relative motion constraints, without joint optimisation or centralised decision-making.

2.4 Path and Trajectory Planning

Path and trajectory planning address the problem of generating feasible, collision-free, and dynamically consistent motions for autonomous systems operating in structured or unstructured environments, translating abstract goals into trackable trajectories.

2.4.1 Offline and Online Planning

Methods are broadly divided into offline and online approaches, depending on when the plan is computed and how environmental information is handled:

- **Offline Planning:** Performed before execution, assuming full or near-complete knowledge of the operating environment. Classical approaches include graph-search methods (Dijkstra's, A*), sampling-based methods (RRT, PRM), and trajectory optimization. While generating globally optimal paths, they are computationally demanding and cannot adapt to dynamic changes.

- **Online Planning:** Addresses the generation and adaptation of trajectories in real time under dynamic or uncertain conditions. Reactive techniques such as potential field methods and dynamic window approaches provide fast obstacle avoidance but can suffer from local minima. Optimization-based MPC frameworks integrate vehicle dynamics into the planning loop, generating dynamically consistent collision-free paths online.

2.4.2 Related Work on Trajectory Planning

A centralised predictive planning approach is presented in [14], where a single MPC formulation jointly optimises the motion of both the UAV and the USV. In this case, planning is achieved online through joint trajectory optimisation that explicitly accounts for the coupling introduced by tethers.

Planning is addressed from a different perspective in [15], where cooperative behaviour is achieved through coordinated path following. A predefined path is assigned to the USV, while the UAV generates its motion online as part of the control process to maintain coordination. In this framework, planning is limited to the online generation of the UAV motion, and no cooperative path planning between the two vehicles is performed.

Overall, existing works typically adopt one of two approaches to planning in UAV–USV systems: either a predefined path is assigned to one vehicle while the other generates its motion online to follow or support it, or both vehicles generate their motion online as part of a predictive control framework.

Chapter 3

Proposed Methodology

This chapter presents the theoretical and algorithmic methodology developed in this work for the autonomous cooperative control of the tethered USV-UAV system. It is structured into two main parts: Section 3.1 derives the control-oriented kinematic and dynamic models of both vehicles, their actuator mappings, and the physical coupling introduced by the tether; Section 3.2 then formulates the cooperative Nonlinear Model Predictive Control (NMPC) architecture that exploits these models to compute optimal, collision-free, and constraint-respecting trajectories.

3.1 System Modeling

Real-time control needs models that are simple enough to compute quickly, yet detailed enough to capture how each vehicle actually moves and interacts with the other. The kinematics, dynamics, and physical coupling between the two vehicles let the controller predict how the system evolves and compute feasible, optimal trajectories.

The reference frames and notation are fixed first, followed by the kinematic and dynamic models of the USV and the UAV.

3.1.1 Reference Coordinate Frames and Notation

Deriving the equations of motion requires specifying the frames against which positions, orientations, and velocities are measured. Since the vessel and the drone move both with respect to the world and with respect to each other, a common frame of reference is required to describe their motion. Figure 3.1 illustrates these frames.

Inertial Reference Frame $\{\mathcal{I}\}$ This frame establishes the global coordinate system following the East-North-Up (ENU) convention, aligned with the standard ROS convention. Its origin $\mathcal{O}_I$ is fixed at an arbitrary point on the water surface within the operating area, following the usual flat-Earth convention. The axis $\mathbf{x}^{\mathcal{I}}$ points East, $\mathbf{y}^{\mathcal{I}}$ points North, and $\mathbf{z}^{\mathcal{I}}$ points upward, opposite to gravity.

USV Body-Fixed Reference Frame $\{\mathcal{B}_v\}$ This frame is rigidly attached to the vessel, with origin $\mathcal{O}_v$ at its centre of gravity. This choice decouples the linear and rotational momentum equations. The forward-pointing axis $\mathbf{x}^{\mathcal{B}_v}$ defines surge, $\mathbf{y}^{\mathcal{B}_v}$ points to port (sway), and $\mathbf{z}^{\mathcal{B}_v}$ points upward (heave).

UAV Body-Fixed Reference Frame $\{\mathcal{B}_a\}$ Similarly, this frame attaches to the UAV, with origin $\mathcal{O}_a$ at its centre of gravity, decoupling the translational and rotational dynamics in the same way. The longitudinal axis $\mathbf{x}^{\mathcal{B}_a}$ points forward, the lateral axis $\mathbf{y}^{\mathcal{B}_a}$ points to the left, and the vertical axis $\mathbf{z}^{\mathcal{B}_a}$ points upward, aligning with the collective thrust direction.

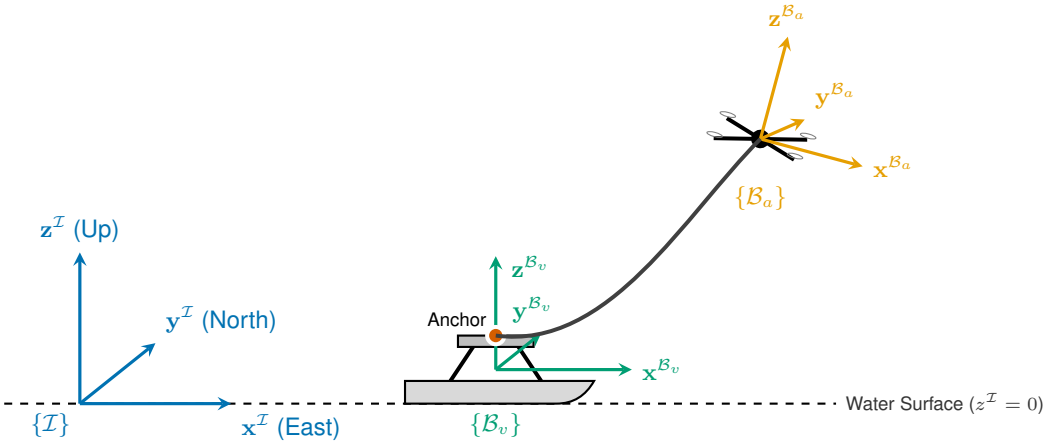


Figure 3.1: Reference coordinate frames $\{\mathcal{I}\}$, $\{\mathcal{B}_v\}$, and $\{\mathcal{B}_a\}$.

3.1.2 WAM-V Surface Vessel Model

The mathematical model of the WAM-V catamaran, covering its horizontal maneuvering dynamics and thruster allocation, forms a control-oriented state-space model suitable for trajectory tracking and predictive control.

Although a rigid body moving freely in the marine environment inherently has six degrees of freedom, the WAM-V is a vessel that moves on the surface, which allows the motion to be simplified to the horizontal plane. Under this assumption, the model is reduced to three degrees of freedom.

Predicting the vessel's motion, as needed for trajectory tracking and predictive control, requires knowing how its state changes over time for a given input. This relationship is captured by the nonlinear state-space model

$$\dot{\mathbf{x}}_v = f(\mathbf{x}_v, \mathbf{u}_v), \tag{3.1}$$

The state vector $\mathbf{x}_v \in \mathbb{R}^6$ is given by

$$\mathbf{x}_v = [x, y, \psi, u, v, r]^\top, \tag{3.2}$$

where:

- x, y – position of the origin $\mathcal{O}_v$ of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;
- ψ – heading of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$;

- u, v – surge and sway linear velocities of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{B}_v\}$;
- r – angular velocity of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{B}_v\}$;

The control input vector $\mathbf{u}_v \in \mathbb{R}^3$ is given by

$$\mathbf{u}_v = [X, Y, N]^\top, \tag{3.3}$$

where:

- X – surge force generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;
- Y – sway force generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;
- N – yaw moment generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;

The vector field $f(\cdot)$ splits into a kinematic block and a dynamic block, derived in turn in the following subsections.

Kinematic Model

The kinematics describe how the vessel's inertial position and heading evolve as a function of its body-fixed velocities, forming the first half of (3.1).

Let $\mathbf{p} = x\,\mathbf{x}^{\mathcal{I}} + y\,\mathbf{y}^{\mathcal{I}}$ denote the inertial position of the origin $\mathcal{O}_v$ of $\{\mathcal{B}_v\}$. Since $\mathbf{x}^{\mathcal{I}}, \mathbf{y}^{\mathcal{I}}$ are constant, differentiating yields

$$\dot{\mathbf{p}} = \dot{x}\,\mathbf{x}^{\mathcal{I}} + \dot{y}\,\mathbf{y}^{\mathcal{I}}, \tag{3.4}$$

i.e. $\dot{x}, \dot{y}$ are, by definition, the components of $\dot{\mathbf{p}}$ in the inertial frame.

The surge and sway velocities u, v are defined in the body-fixed frame, so that

$$\dot{\mathbf{p}} = u\,\mathbf{x}^{\mathcal{B}} + v\,\mathbf{y}^{\mathcal{B}}, \tag{3.5}$$

where $\mathbf{x}^{\mathcal{B}}, \mathbf{y}^{\mathcal{B}}$ denote the basis vectors of $\{\mathcal{B}_v\}$, i.e. $\mathbf{x}^{\mathcal{B}_v}, \mathbf{y}^{\mathcal{B}_v}$.

Since $\{\mathcal{B}_v\}$ is rotated relative to $\{\mathcal{I}\}$ by the heading angle ψ, the inertial unit vectors are related to the body-fixed ones by the rotation matrix $\mathbf{R}(\psi) \in SO(2)$,

$$\begin{bmatrix} \mathbf{x}^{\mathcal{I}} \\ \mathbf{y}^{\mathcal{I}} \end{bmatrix} = \mathbf{R}(\psi) \begin{bmatrix} \mathbf{x}^{\mathcal{B}} \\ \mathbf{y}^{\mathcal{B}} \end{bmatrix}, \tag{3.6}$$

following the standard planar rotation matrix,

$$\mathbf{R}(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}, \tag{3.7}$$

which is equivalent to

$$\mathbf{x}^{\mathcal{B}} = \cos\psi\,\mathbf{x}^{\mathcal{I}} + \sin\psi\,\mathbf{y}^{\mathcal{I}}, \qquad \mathbf{y}^{\mathcal{B}} = -\sin\psi\,\mathbf{x}^{\mathcal{I}} + \cos\psi\,\mathbf{y}^{\mathcal{I}}. \tag{3.8}$$

Substituting (3.8) into (3.5) and grouping terms gives

$$\dot{\mathbf{p}} = \underbrace{(u\cos\psi - v\sin\psi)}_{\dot{x}}\,\mathbf{x}^{\mathcal{I}} + \underbrace{(u\sin\psi + v\cos\psi)}_{\dot{y}}\,\mathbf{y}^{\mathcal{I}}. \tag{3.9}$$

Comparing with (3.4) then directly gives

$$\dot{x} = u\cos\psi - v\sin\psi, \qquad \dot{y} = u\sin\psi + v\cos\psi, \tag{3.10}$$

which is precisely

$$\begin{bmatrix}\dot{x}\\ \dot{y}\end{bmatrix} = \mathbf{R}(\psi)\begin{bmatrix}u\\ v\end{bmatrix}. \tag{3.11}$$

The heading rate follows directly from the definition of the angular velocity r:

$$\dot{\psi} = r. \tag{3.12}$$

Dynamic Model

The dynamics describe how the vessel's body-fixed velocities evolve as a function of the applied forces and moments, forming the second half of (3.1).

Unlike $\mathbf{x}^{\mathcal{I}}, \mathbf{y}^{\mathcal{I}}$, the body-fixed unit vectors $\mathbf{x}^{\mathcal{B}}, \mathbf{y}^{\mathcal{B}}$ rotate with the vehicle. Differentiating (3.8) and using $\dot{\psi} = r$ gives

$$\dot{\mathbf{x}}^{\mathcal{B}} = r\,\mathbf{y}^{\mathcal{B}}, \qquad \dot{\mathbf{y}}^{\mathcal{B}} = -r\,\mathbf{x}^{\mathcal{B}}. \tag{3.13}$$

The inertial velocity, expressed in the body-fixed frame, is $\dot{\mathbf{p}} = u\,\mathbf{x}^{\mathcal{B}} + v\,\mathbf{y}^{\mathcal{B}}$. Differentiating once more and substituting (3.13) yields the inertial acceleration

$$\ddot{\mathbf{p}} = (\dot{u} - vr)\,\mathbf{x}^{\mathcal{B}} + (\dot{v} + ur)\,\mathbf{y}^{\mathcal{B}}, \tag{3.14}$$

where the cross terms $-vr$ and ur are Coriolis/centripetal terms arising because $\{\mathcal{B}_v\}$ is a rotating, non-inertial frame.

Let F_x, F_y denote the resultant surge and sway forces applied at $\mathcal{O}_v$, so that $\mathbf{F} = F_x\,\mathbf{x}^{\mathcal{B}} + F_y\,\mathbf{y}^{\mathcal{B}}$. By Newton's second law, $\mathbf{F} = m\,\ddot{\mathbf{p}}$, and comparing with (3.14),

$$F_x = m(\dot{u} - vr), \qquad F_y = m(\dot{v} + ur). \tag{3.15}$$

Since the vessel's motion is planar, Euler's rotational equation about $\mathbf{z}^{\mathcal{B}}$ takes the form

$$M_z = I_z\,\dot{r}, \tag{3.16}$$

where M_z is the resultant yaw moment about $\mathcal{O}_v$ and I_z the yaw moment of inertia.

The resultant force and moment F_x, F_y, M_z combine two contributions: the thrust generated by the

propulsion system, X, Y, N, and the hydrodynamic damping opposing the vessel's motion. This damping follows the standard linear-plus-quadratic model for surface vessels [22]. The quadratic surge term is further split into forward and backward coefficients, to capture the asymmetric drag between the two directions:

$$F_x = X - \big(X_u + X_{uu}^{\pm}|u|\big)u, \qquad X_{uu}^{\pm} = \begin{cases} X_{uu}^{+}, & u \geq 0 \\ X_{uu}^{-}, & u < 0 \end{cases} \tag{3.17}$$

$$F_y = Y - \big(Y_v + Y_{vv}|v|\big)v, \qquad M_z = N - \big(N_r + N_{rr}|r|\big)r. \tag{3.18}$$

Substituting (3.17)–(3.18) into (3.15) and (3.16), and solving for the accelerations, gives the dynamic block of (3.1):

$$\dot{u} = \frac{1}{m}\Big[X - \big(X_u + X_{uu}^{\pm}|u|\big)u\Big] + vr, \tag{3.19}$$

$$\dot{v} = \frac{1}{m}\Big[Y - \big(Y_v + Y_{vv}|v|\big)v\Big] - ur, \tag{3.20}$$

$$\dot{r} = \frac{1}{I_z}\Big[N - \big(N_r + N_{rr}|r|\big)r\Big]. \tag{3.21}$$

Thruster Allocation

The thruster allocation maps the physical actuator commands, the left and right thrust magnitudes T_L, T_R and azimuth angles α_L, α_R, to the resultant force and moment X, Y, N required by the dynamics of Section 3.1.2, according to the geometry shown in Figure 3.2.

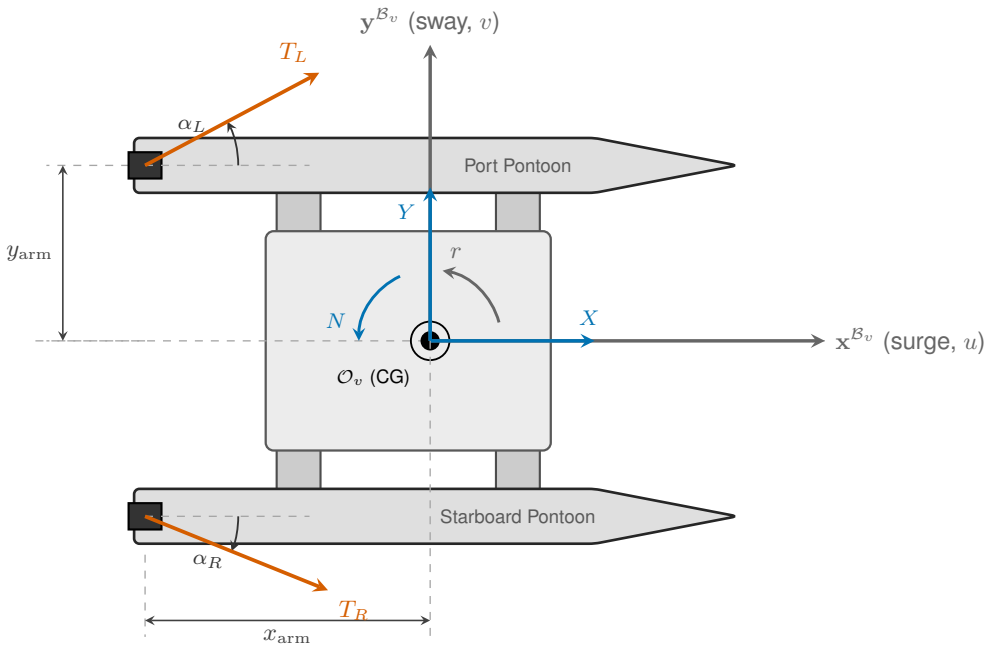


Figure 3.2: Planar USV schematic and thruster allocation geometry.

Each azimuth angle is measured relative to $\mathbf{x}^{\mathcal{B}}$, so that each thruster produces a planar force

$$\mathbf{f}_L = T_L \cos\alpha_L\, \mathbf{x}^{\mathcal{B}} + T_L \sin\alpha_L\, \mathbf{y}^{\mathcal{B}}, \qquad \mathbf{f}_R = T_R \cos\alpha_R\, \mathbf{x}^{\mathcal{B}} + T_R \sin\alpha_R\, \mathbf{y}^{\mathcal{B}}. \tag{3.22}$$

Summing the components of both thrusters along $\mathbf{x}^{\mathcal{B}}$ and $\mathbf{y}^{\mathcal{B}}$ gives the resultant force applied at $\mathcal{O}_v$:

$$X = T_L \cos\alpha_L + T_R \cos\alpha_R, \qquad Y = T_L \sin\alpha_L + T_R \sin\alpha_R. \tag{3.23}$$

Both thrusters are mounted behind $\mathcal{O}_v$, at the same longitudinal distance x_{arm}, and symmetrically on either side of the centerline, at a lateral distance y_{arm}. Expressed in $\{\mathcal{B}_v\}$, the left (port) thruster is located at $(-x_{\mathrm{arm}}, y_{\mathrm{arm}})$ and the right (starboard) thruster at $(-x_{\mathrm{arm}}, -y_{\mathrm{arm}})$.

The yaw moment produced by a planar force (f_x, f_y) applied at (r_x, r_y) is given by the two-dimensional cross product $r_x f_y - r_y f_x$. Applying this to each thruster and summing gives the resultant yaw moment about $\mathcal{O}_v$:

$$N = -x_{\mathrm{arm}}\big(T_L \sin\alpha_L + T_R \sin\alpha_R\big) - y_{\mathrm{arm}}\big(T_L \cos\alpha_L - T_R \cos\alpha_R\big). \tag{3.24}$$

Equations (3.23) and (3.24) complete the mapping from the four physical actuator commands $T_L, T_R, \alpha_L, \alpha_R$ to the control input $\mathbf{u}_v = [X, Y, N]^\top$ used by the dynamic model.

3.1.3 x500 Quadrotor Model

The mathematical model of the x500 quadrotor, covering its translational dynamics in three-dimensional space, forms a control-oriented state-space model suitable for trajectory tracking and predictive control. The vehicle's attitude is stabilized by an inner-loop attitude controller, to which the attitude used in the translational dynamics is commanded.

Predicting the vehicle's motion, as needed for trajectory tracking and predictive control, requires knowing how its state changes over time for a given input. This relationship is captured by the nonlinear state-space model

$$\dot{\mathbf{x}}_a = f(\mathbf{x}_a, \mathbf{u}_a), \tag{3.25}$$

The state vector $\mathbf{x}_a \in \mathbb{R}^6$ is given by

$$\mathbf{x}_a = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top, \tag{3.26}$$

where:

- x, y, z – position of the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;
- $\dot{x}, \dot{y}, \dot{z}$ – linear velocity of $\mathcal{O}_a$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;

The control input vector $\mathbf{u}_a \in \mathbb{R}^4$ is given by

$$\mathbf{u}_a = [T, \phi, \theta, \psi]^\top, \tag{3.27}$$

where:

- T – total thrust force, applied to $\{\mathcal{B}_a\}$, expressed in $\{\mathcal{B}_a\}$;
- ϕ, θ, ψ – roll, pitch, and yaw angles of $\{\mathcal{B}_a\}$ relative to $\{\mathcal{I}\}$;

The vector field $f(\cdot)$ splits into a kinematic block, relating the inertial position and velocity, and a dynamic block, mapping the thrust, attitude, aerodynamic drag, and tether force to the inertial translational acceleration.

Kinematic Model

The kinematics describe how the vehicle's inertial position evolves as a function of its inertial velocity, forming the first half of (3.25).

Let $\mathbf{p} = x\,\mathbf{x}^{\mathcal{I}} + y\,\mathbf{y}^{\mathcal{I}} + z\,\mathbf{z}^{\mathcal{I}}$ denote the inertial position of the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$. Since $\mathbf{x}^{\mathcal{I}}, \mathbf{y}^{\mathcal{I}}, \mathbf{z}^{\mathcal{I}}$ are constant, differentiating yields

$$\dot{\mathbf{p}} = \dot{x}\,\mathbf{x}^{\mathcal{I}} + \dot{y}\,\mathbf{y}^{\mathcal{I}} + \dot{z}\,\mathbf{z}^{\mathcal{I}}, \tag{3.28}$$

i.e. $\dot{x}, \dot{y}, \dot{z}$ are, by definition, the components of $\dot{\mathbf{p}}$ in the inertial frame, which are precisely the linear velocity components in $\mathbf{x}_a$. The kinematic block of (3.25) is therefore the trivial identity

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}. \tag{3.29}$$

This is in contrast to a surface vessel such as the WAM-V (Section 3.1.2), where the linear velocities are instead expressed in $\{\mathcal{B}_v\}$, requiring an additional change of basis via $\mathbf{R}(\psi)$ to relate them to $\dot{\mathbf{p}}$. Here, this rotation is required instead to express the thrust force, applied along $\mathbf{z}^{\mathcal{B}_a}$, in $\{\mathcal{I}\}$. It is therefore deferred to the dynamic block, derived next.

Dynamic Model

The dynamics describe how the vehicle's inertial velocity evolves as a function of the applied thrust, attitude, aerodynamic drag, and tether force, forming the second half of (3.25).

The thrust generated by the propulsion system acts along the vertical axis of $\{\mathcal{B}_a\}$, so that the resultant thrust force is

$$\mathbf{F}_{\text{thr}} = T\,\mathbf{z}^{\mathcal{B}}, \tag{3.30}$$

where $\mathbf{z}^{\mathcal{B}}$ denotes the vertical basis vector of $\{\mathcal{B}_a\}$, i.e. $\mathbf{z}^{\mathcal{B}_a}$.

Since (3.29) relates the inertial position and velocity directly, without an intermediate change of basis, Newton's second law is more conveniently applied directly in $\{\mathcal{I}\}$, requiring $\mathbf{F}_{\text{thr}}$ to instead be expressed in the inertial frame.

Since $\{\mathcal{B}_a\}$ is rotated relative to $\{\mathcal{I}\}$ by the roll, pitch, and yaw angles ϕ, θ, ψ, the inertial unit vectors are related to the body-fixed ones by the rotation matrix $\mathbf{R}(\phi, \theta, \psi) \in SO(3)$,

$$\begin{bmatrix} \mathbf{x}^{\mathcal{I}} \\ \mathbf{y}^{\mathcal{I}} \\ \mathbf{z}^{\mathcal{I}} \end{bmatrix} = \mathbf{R}(\phi, \theta, \psi) \begin{bmatrix} \mathbf{x}^{\mathcal{B}} \\ \mathbf{y}^{\mathcal{B}} \\ \mathbf{z}^{\mathcal{B}} \end{bmatrix}, \tag{3.31}$$

following the ZYX (yaw-pitch-roll) Euler angle convention,

$$\mathbf{R}(\phi,\theta,\psi) = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}, \tag{3.32}$$

where $s_\bullet = \sin(\bullet)$ and $c_\bullet = \cos(\bullet)$.

Projecting $\mathbf{F}_{\text{thr}}$ onto $\{\mathcal{I}\}$ then gives

$$\mathbf{F}^{\mathcal{I}}_{\text{thr}} = \mathbf{R}(\phi,\theta,\psi) \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}. \tag{3.33}$$

Figure 3.3 illustrates this projection in the longitudinal plane, showing how the body tilt θ splits the thrust force T into vertical and horizontal inertial components.

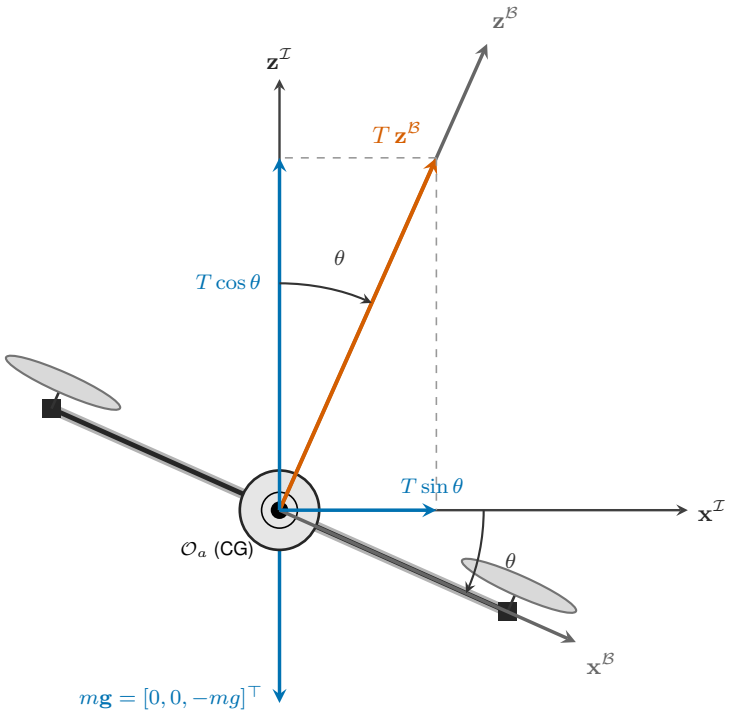


Figure 3.3: Quadrotor thrust vectoring in the longitudinal plane.

Gravity, unlike the thrust, is already naturally expressed in $\{\mathcal{I}\}$:

$$\mathbf{F}^{\mathcal{I}}_{\text{grav}} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}. \tag{3.34}$$

During translational flight, the vehicle is also subject to aerodynamic drag opposing its planar motion, arising primarily from the relative airflow across the spinning rotor disks (rotor drag). This force is natu-

rally expressed as a linear damping of the velocity in $\{B_a\}$, with coefficient d_r along the body axes x^B and y^B. For small roll and pitch angles, the horizontal axes of the body remain approximately horizontal in $\{I\}$, so the drag can be written directly in the inertial frame as

$$\mathbf{F}^{\mathcal{I}}_{\text{drag}} = \begin{bmatrix} -d_r\,\dot{x} \\ -d_r\,\dot{y} \\ 0 \end{bmatrix}, \tag{3.35}$$

where the small coupling between horizontal and vertical velocities caused by the tilt of the vehicle is neglected.

In addition to thrust, gravity, and aerodynamic drag, the vehicle is subject to a force transmitted through a tether attached at the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$, denoted $\mathbf{F}^{\mathcal{I}}_{\text{tether}}$ and derived separately in Section 3.1.3.

Since $\{\mathcal{I}\}$ is non-rotating, Newton's second law applies directly to the inertial acceleration, without the transport terms that arise when applying it in a rotating frame, such as $\{\mathcal{B}_v\}$ in the WAM-V model (Section 3.1.2):

$$m\,\ddot{\mathbf{p}} = \mathbf{F}^{\mathcal{I}}_{\text{thr}} + \mathbf{F}^{\mathcal{I}}_{\text{grav}} + \mathbf{F}^{\mathcal{I}}_{\text{drag}} + \mathbf{F}^{\mathcal{I}}_{\text{tether}}. \tag{3.36}$$

Substituting (3.33), (3.34), and (3.35) into (3.36), and solving for the acceleration, gives the dynamic block of (3.25):

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \frac{T}{m}\,\mathbf{R}(\phi,\theta,\psi) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} - \frac{d_r}{m} \begin{bmatrix} \dot{x} \\ \dot{y} \\ 0 \end{bmatrix} + \frac{1}{m}\mathbf{F}^{\mathcal{I}}_{\text{tether}}. \tag{3.37}$$

Tether Force Model

In addition to thrust and gravity, the vehicle is subject to a force transmitted through a tether connecting the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$ to a fixed attachment point on the WAM-V, whose position $\mathbf{p}_v$ is treated here only as a known geometric anchor, without modeling the (negligible) reaction force exerted on the boat.

The relevant geometric quantity is the straight-line vector between the two attachment points,

$$\mathbf{r} = \mathbf{p}_v - \mathbf{p}, \qquad d = \|\mathbf{r}\|, \qquad \hat{\mathbf{d}}_0 = \frac{\mathbf{r}}{d}, \tag{3.38}$$

where d is the straight-line distance between the vehicle and the boat, and $\hat{\mathbf{d}}_0$ is the direction the tether would have if taut.

Denoting by $L \geq d$ the tether length at a given instant, its slack is conveniently quantified by the dimensionless ratio

$$\varepsilon = \frac{L - d}{L}, \tag{3.39}$$

with $\varepsilon = 0$ corresponding to a taut tether, and ε increasing as more slack is introduced.

Under its own weight, a tether of linear mass density μ suspended between two attachment points takes the shape of a catenary, contained in the vertical plane defined by the two endpoints. Due to

the geometric self-similarity of the catenary curve at a given slack ratio ε, the tension magnitude scales proportionally with the straight-line separation distance d (or unstretched length L), while the tension direction is governed by the angular deflection γ by which the tension vector at $\mathcal{O}_a$ deviates from $\hat{\mathbf{d}}_0$ towards the vertical.

The exact relations governing the deflection angle $\gamma_{\text{exact}}(\varepsilon, \alpha)$ and the normalized tension coefficient $c_d^{\text{exact}}(\varepsilon, \alpha) = T_{\text{tether}}/d$ do not admit a closed-form analytical solution, requiring numerical root-finding algorithms at every control cycle.

The tether control scheme considered in this work maintains ε close to a constant operating point ε_0. Under this operational regime, the tension magnitude scales linearly with the separation distance d:

$$T_{\text{tether}}(d) = c_d(\varepsilon_0)\, d, \tag{3.40}$$

where the nominal coefficient at ε_0 is evaluated from first principles:

$$c_d(\varepsilon_0) = \frac{\mu\, g}{1 - \varepsilon_0}, \tag{3.41}$$

or equivalently in terms of unstretched length, $T_{\text{tether}}(L) = \mu\, g\, L$. Similarly, the catenary deflection angle is given by its nominal value at ε_0:

$$\gamma = \gamma(\varepsilon_0), \tag{3.42}$$

which replaces the transcendental relation with a closed-form expression evaluated at ε_0.

With γ given by (3.42), the tension direction $\hat{\mathbf{d}}$ can now be constructed. This requires first introducing the direction $\hat{\mathbf{e}}_h$, orthogonal to $\mathbf{z}^{\mathcal{I}}$ within the catenary plane,

$$\hat{\mathbf{e}}_h = \frac{\hat{\mathbf{d}}_0 - (\hat{\mathbf{d}}_0 \cdot \mathbf{z}^{\mathcal{I}})\, \mathbf{z}^{\mathcal{I}}}{\|\hat{\mathbf{d}}_0 - (\hat{\mathbf{d}}_0 \cdot \mathbf{z}^{\mathcal{I}})\, \mathbf{z}^{\mathcal{I}}\|}, \tag{3.43}$$

and the depression angle α of $\hat{\mathbf{d}}_0$ below $\hat{\mathbf{e}}_h$, which follows from

$$\hat{\mathbf{d}}_0 = \cos\alpha\, \hat{\mathbf{e}}_h - \sin\alpha\, \mathbf{z}^{\mathcal{I}}. \tag{3.44}$$

Figure 3.4 illustrates this construction within the catenary plane, showing the local horizontal $\hat{\mathbf{e}}_h$, the depression angle α, and the resulting tension direction $\hat{\mathbf{d}}$, tilted further from $\hat{\mathbf{d}}_0$ by the angle γ.

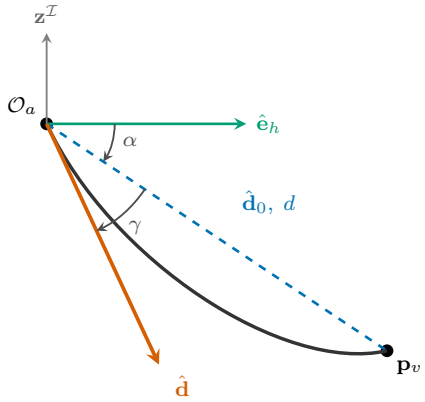


Figure 3.4: Tether catenary plane.

The tension direction at $\mathcal{O}_a$ then follows by tilting $\hat{\mathbf{d}}_0$ further towards the vertical by the angle γ, within the same plane,

$$\hat{\mathbf{d}} = \cos(\alpha + \gamma)\,\hat{\mathbf{e}}_h - \sin(\alpha + \gamma)\,\mathbf{z}^{\mathcal{I}}, \tag{3.45}$$

and the resulting tether force, applied at $\mathcal{O}_a$ and expressed in $\{\mathcal{I}\}$, is

$$\mathbf{F}^{\mathcal{I}}_{\text{tether}} = T_{\text{tether}}(d)\,\hat{\mathbf{d}} = c_d(\varepsilon_0)\,d\,\hat{\mathbf{d}}, \tag{3.46}$$

with d evaluated at each instant from (3.38). Equation (3.46) is the term $\mathbf{F}^{\mathcal{I}}_{\text{tether}}$ introduced in (3.36).

3.2 System Control

Having established, in Chapter 3.1, the kinematic and dynamic models governing the USV and UAV, the natural next step is to determine how these models can be exploited to compute the control inputs that drive each vehicle along a desired trajectory. Because the two vehicles are physically coupled through the tether, their motion cannot be planned independently without accounting for this interaction, motivating a cooperative control strategy.

The overall control approach is therefore established first, followed by the design of the controllers for each vehicle.

3.2.1 Cooperative Control Design

Before formulating either controller, the strategies available are weighed against the system's dynamics and its objectives, so that the method chosen suits both vehicles.

Control Method

Keeping the tether within safe operating conditions depends on how the distance between the two vehicles evolves over time. A controller that only reacts to the current state can correct a violation only

after it happens. Anticipating where the system is headed allows corrective action before reaching the safety limits.

A predictive controller does this by repeatedly solving, at each control instant, an optimization problem over a finite look-ahead window. Because the optimization runs over the predicted trajectory rather than the current state alone, requirements such as keeping the tether within safe conditions can be stated as explicit constraints in the problem itself, enforcing them explicitly over the prediction horizon.

The predictions rely on the model of the system dynamics, so their accuracy is tied to how well that model represents the real behaviour. The USV and UAV models derived in Chapter 3.1 include nonlinear terms. The predictive controller uses these nonlinear models directly, making non-linear model predictive control the method used for both vehicles.

Cooperative Architecture

The USV and the UAV operate on different timescales. The USV moves in the horizontal plane with slow, inertia-dominated motion. The UAV's dynamics evolve much faster and demand a finer sampling time. A single NMPC built around the combined state of both vehicles would force one prediction horizon and one sampling time onto both, compromising one vehicle's timescale to accommodate the other.

Combining both vehicles into one NMPC would also enlarge the state, input, and constraint sets a single solver must handle, making a problem that already runs under real-time constraints harder to solve within the available computation time. Keeping the two problems separate matches each optimization's size to its own vehicle and keeps the computation more manageable in real time.

Each vehicle is therefore controlled by its own NMPC. This separation isolates the two controllers from each other's failures: if one solver fails to find a feasible solution, the other vehicle's NMPC continues operating on its own well-posed problem.

The tether's safety limits depend on the relative position between the USV and the UAV, so each vehicle's NMPC accounts for the other vehicle's position. At every control instant, each controller receives the other vehicle's most recently computed predicted trajectory, and uses it to compute the predicted inter-vehicle distance at the nodes of its own horizon covered by that trajectory. Because the two controllers operate at different sampling intervals, the received predicted trajectory is temporally interpolated and resampled at the receiver's own prediction nodes.

Without this exchange, a vehicle would have to treat the other as stationary or extrapolate its motion from the last known state. Either assumption misrepresents the tether coupling and can drive the system into a constraint violation.

Because the UAV operates with a finer sampling time and a possibly shorter prediction horizon than the USV, its predicted trajectory may cover only part of the USV's horizon. The USV therefore enforces the tether constraint only at the nodes covered by the received trajectory, and leaves it inactive at the remaining ones, where no information about the UAV is available. This avoids constraining the USV with an assumed UAV position that has no basis in the prediction. Since the problem is solved again at every instant, the covered window slides forward with time, and any potential violation is accounted for as soon as it enters it. In the opposite direction no such restriction is needed, since the USV's horizon

fully covers that of the UAV.

Figure 3.5 shows this architecture, with two independent NMPC loops, each driving its own vehicle, exchanging predicted horizons to enforce the tether constraint.

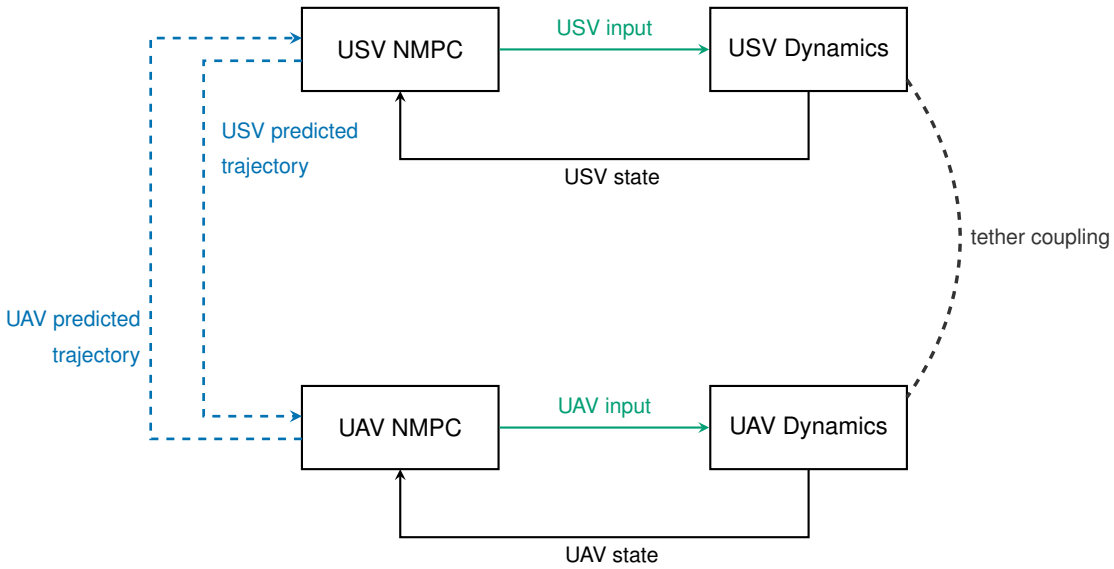


Figure 3.5: Cooperative control architecture diagram.

3.2.2 Nonlinear Model Predictive Control

The vehicle models of Chapter 3.1 are continuous-time systems of the form

$$\dot{\mathbf{x}} = \mathbf{f}\left(\mathbf{x}, \mathbf{u}\right), \tag{3.47}$$

where $\mathbf{x} \in \mathbb{R}^{n_x}$ is the state vector and $\mathbf{u} \in \mathbb{R}^{n_u}$ is the control input vector.

However, the controller acts at discrete sampling instants $t_k = kT_s$, with the input held constant over each interval $[t_k, t_{k+1})$. Under this assumption, the state at the next instant follows from integrating the dynamics over one sampling interval,

$$\mathbf{x}(t_{k+1}) = \mathbf{x}(t_k) + \int_{t_k}^{t_{k+1}} \mathbf{f}\left(\mathbf{x}(\tau), \mathbf{u}_k\right) d\tau. \tag{3.48}$$

Since this integral generally has no closed-form solution for nonlinear dynamics, it is approximated by applying a numerical integration scheme to $\mathbf{f}$ over one sampling interval. This defines a one-step discrete map $\mathbf{\Phi}$,

$$\mathbf{x}_{k+1} = \mathbf{\Phi}\left(\mathbf{x}_k, \mathbf{u}_k\right), \tag{3.49}$$

where $\mathbf{x}_k \approx \mathbf{x}(t_k)$ is the system state at instant k, so that $\mathbf{\Phi}$ returns the approximate state one sampling interval ahead, given the current state and input.

At every instant k, NMPC uses this map to predict the future behaviour of the system over a finite look-ahead window of N steps, and computes the input that steers it toward a reference. Denote by $\hat{\mathbf{x}}_{k+i|k}$ and $\hat{\mathbf{u}}_{k+i|k}$ the state and input predicted i steps ahead at instant k, for $i = 0, \ldots, N$ and $i = 0, \ldots, N-1$,

respectively. The predicted trajectory starts from the estimated current state,

$$\hat{\mathbf{x}}_{k|k} = \mathbf{x}_k, \tag{3.50}$$

and evolves according to the system model,

$$\hat{\mathbf{x}}_{k+i+1|k} = \mathbf{\Phi}\left(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}\right), \quad i = 0, \ldots, N-1. \tag{3.51}$$

Each predicted step is then scored against a reference trajectory. A stage cost $\ell(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k})$, typically quadratic, penalizes the tracking error and the control effort at every step of the horizon, and a terminal cost $\ell_f(\hat{\mathbf{x}}_{k+N|k})$ penalizes the tracking error at its final step. Summing both over the horizon gives the total predicted cost,

$$J = \sum_{i=0}^{N-1} \ell\left(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}\right) + \ell_f\left(\hat{\mathbf{x}}_{k+N|k}\right). \tag{3.52}$$

Minimizing J alone ignores the physical limits of the system, so the prediction is also constrained. The predicted states and inputs must remain within the admissible sets $\mathcal{X}$ and $\mathcal{U}$ at every step of the horizon,

$$\hat{\mathbf{x}}_{k+i|k} \in \mathcal{X}, \quad i = 0, \ldots, N, \tag{3.53}$$

$$\hat{\mathbf{u}}_{k+i|k} \in \mathcal{U}, \quad i = 0, \ldots, N-1, \tag{3.54}$$

where $\mathcal{X}$ and $\mathcal{U}$ collect the admissible states and inputs. These sets may be defined by simple bounds or by general nonlinear inequality constraints of the form $h(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) \leq 0$.

Enforcing these sets as hard constraints can make the problem infeasible. If a disturbance pushes the true state outside the admissible set, no predicted trajectory satisfies the constraints, and the solver has no solution to return even though one is required at every sampling instant. To avoid this, a constraint $h(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) \leq 0$ can be softened by a slack variable $\sigma \geq 0$,

$$h(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) \leq \sigma, \tag{3.55}$$

which allows the constraint to be violated by an amount σ. The slack becomes an additional decision variable at each step where the constraint is softened, and a penalty

$$\rho(\sigma) = \mu_1 \sigma + \mu_2 \sigma^2, \qquad \mu_1, \mu_2 \geq 0, \tag{3.56}$$

is added to the cost J for each slack. Since $\rho(0) = 0$, the solver prefers $\sigma = 0$ whenever the original constraint can be met, and large weights μ_1, μ_2 keep any violation close to zero.

Assembling the cost, the prediction model, and the constraints defines the optimal control problem

solved at every sampling instant k,

$$\min_{\hat{\mathbf{x}}_{k+i|k},\, \hat{\mathbf{u}}_{k+i|k},\, \hat{\sigma}_{k+i|k}} \quad J + \sum_{i=0}^{N-1} \rho\left(\hat{\sigma}_{k+i|k}\right) \tag{3.57}$$

$$\begin{aligned}
\text{s.t.} \quad & \hat{\mathbf{x}}_{k|k} = \mathbf{x}_k, \\
& \hat{\mathbf{x}}_{k+i+1|k} = \boldsymbol{\Phi}\left(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}\right), \quad i = 0, \ldots, N-1, \\
& \hat{\mathbf{x}}_{k+i|k} \in \mathcal{X}, \quad i = 0, \ldots, N, \\
& \hat{\mathbf{u}}_{k+i|k} \in \mathcal{U}, \quad i = 0, \ldots, N-1, \\
& h\left(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}\right) \leq \hat{\sigma}_{k+i|k}, \quad \hat{\sigma}_{k+i|k} \geq 0, \quad i = 0, \ldots, N-1.
\end{aligned}$$

Solving this problem returns a full predicted input sequence, $\hat{\mathbf{u}}_{k|k}, \ldots, \hat{\mathbf{u}}_{k+N-1|k}$, but only its first element, $\hat{\mathbf{u}}_{k|k}$, is applied to the system. At the next instant, $k+1$, the state is estimated again and the problem is solved again from $\hat{\mathbf{x}}_{k+1|k+1} = \mathbf{x}_{k+1}$, with the horizon shifted one step forward. This strategy, known as the receding horizon principle, computes every input from the latest state estimate, so the controller can correct for disturbances and model errors that the previous prediction did not anticipate.

3.2.3 USV NMPC

The USV NMPC computes, at every sampling instant, the thruster commands that best track a reference trajectory over a finite prediction horizon, subject to the vessel's dynamics, the physical limits of its actuators, and the tether workspace. It predicts the vessel's motion with the model derived in Chapter 3.1, and solves a constrained optimization problem at each step to obtain the commands applied to the thrusters.

Predictive Model

The vessel model of Chapter 3.1 is driven by the resultant thrust force and moment $\boldsymbol{\tau} = [X, Y, N]^\top$, which are obtained from the physical actuator commands $T_L, T_R, \alpha_L, \alpha_R$ through the allocation

$$X = T_L \cos\alpha_L + T_R \cos\alpha_R, \qquad Y = T_L \sin\alpha_L + T_R \sin\alpha_R, \tag{3.58}$$

$$N = -x_{arm}\big(T_L \sin\alpha_L + T_R \sin\alpha_R\big) - y_{arm}\big(T_L \cos\alpha_L - T_R \cos\alpha_R\big). \tag{3.59}$$

Since the controller ultimately has to command $T_L, T_R, \alpha_L, \alpha_R$, the NMPC is formulated directly in terms of these variables, rather than choosing $\boldsymbol{\tau}$ and allocating it separately. This way, the limits of the physical actuators can be imposed directly in the optimization problem.

Each thruster is limited both in the thrust and azimuth it can deliver,

$$T_{L,min} \leq T_L \leq T_{L,max}, \quad T_{R,min} \leq T_R \leq T_{R,max}, \quad \alpha_{L,min} \leq \alpha_L \leq \alpha_{L,max}, \quad \alpha_{R,min} \leq \alpha_R \leq \alpha_{R,max}, \tag{3.60}$$

and in how fast it can change them,

$$|\dot{T}_L| \leq \dot{T}_{L,max}, \quad |\dot{T}_R| \leq \dot{T}_{R,max}, \quad |\dot{\alpha}_L| \leq \dot{\alpha}_{L,max}, \quad |\dot{\alpha}_R| \leq \dot{\alpha}_{R,max}. \tag{3.61}$$

The rate limits act on the time derivatives of the commands, so imposing them directly as bounds requires these derivatives to become the control input,

$$\mathbf{u}_v = [\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R]^\top \in \mathbb{R}^4, \tag{3.62}$$

which in turn makes $T_L, T_R, \alpha_L, \alpha_R$ part of the state vector,

$$\mathbf{x}_v = [x, y, \psi, u, v, r,\ T_L, T_R, \alpha_L, \alpha_R]^\top \in \mathbb{R}^{10}. \tag{3.63}$$

With this choice, the range limits act on the state, and the rate limits act directly on the input.

With $\mathbf{x}_v$ and $\mathbf{u}_v$ defined, the prediction model $\dot{\mathbf{x}}_v = \mathbf{f}_v(\mathbf{x}_v, \mathbf{u}_v)$ is assembled from the vessel kinematics and dynamics.

The kinematics are given by (3.11) and (3.12),

$$\dot{x} = u\cos\psi - v\sin\psi, \qquad \dot{y} = u\sin\psi + v\cos\psi, \qquad \dot{\psi} = r. \tag{3.64}$$

The dynamics follow (3.19)–(3.21), with X, Y, N given by (3.58)–(3.59) in terms of the thruster state,

$$\dot{u} = \frac{1}{m}\Big[X - \big(X_u + X^{\pm}_{uu}|u|\big)u\Big] + vr, \tag{3.65}$$

$$\dot{v} = \frac{1}{m}\Big[Y - \big(Y_v + Y_{vv}|v|\big)v\Big] - ur, \tag{3.66}$$

$$\dot{r} = \frac{1}{I_z}\Big[N - \big(N_r + N_{rr}|r|\big)r\Big]. \tag{3.67}$$

Finally, $\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R$ appear directly as the components of $\mathbf{u}_v$,

$$\dot{T}_L = \dot{T}_L, \qquad \dot{T}_R = \dot{T}_R, \qquad \dot{\alpha}_L = \dot{\alpha}_L, \qquad \dot{\alpha}_R = \dot{\alpha}_R. \tag{3.68}$$

Discretization over a sampling interval T_s, following Section 3.2.2, gives the one-step map $\boldsymbol{\Phi}_v$ such that $\hat{\mathbf{x}}_{v,k+i+1|k} = \boldsymbol{\Phi}_v\big(\hat{\mathbf{x}}_{v,k+i|k}, \hat{\mathbf{u}}_{v,k+i|k}\big)$ approximates the solution of $\dot{\mathbf{x}}_v = \mathbf{f}_v(\mathbf{x}_v, \mathbf{u}_v)$ over one sampling interval.

Cost and Constraints

Let $\mathbf{p} = [x, y]^\top$ denote the position, ψ the heading, and $\boldsymbol{\nu} = [u, v, r]^\top$ the body-fixed velocities, with $\mathbf{p}_{ref}$, ψ_{ref}, and $\boldsymbol{\nu}_{ref}$ the corresponding references from a given trajectory.

Tracking the reference position is penalized quadratically as

$$\|\mathbf{p} - \mathbf{p}_{ref}\|^2_{Q_p}, \tag{3.69}$$

where $\|\boldsymbol{\xi}\|^2_W \coloneqq \boldsymbol{\xi}^\top W \boldsymbol{\xi}$ for a positive semi-definite weighting matrix W, which sets the relative importance of each component of $\boldsymbol{\xi}$ in the cost.

Tracking the heading reference is penalized as

$$q_\psi\big(1 - \cos(\psi - \psi_{ref})\big), \tag{3.70}$$

where $q_\psi \geq 0$ is a scalar weight setting the relative importance of heading tracking.

Both ψ and ψ_{ref} are angles represented within an interval of width 2π, such as $(-\pi, \pi]$. When they lie near opposite ends of this interval, their raw difference $\psi - \psi_{ref}$ approaches $\pm 2\pi$, even though the two headings are then almost identical. As $\cos(\cdot)$ has period 2π, its value approaches one whenever $\psi - \psi_{ref}$ is near zero or $\pm 2\pi$, so the cost (3.70) is near zero in both cases, correctly reflecting that the two headings are almost identical.

Tracking the reference velocity is penalized in the same way as the position,

$$\|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|^2_{Q_\nu}. \tag{3.71}$$

The thruster commands $T_L, T_R, \alpha_L, \alpha_R$ are regularized directly as

$$\big\|[T_L, T_R, \alpha_L, \alpha_R]^\top\big\|^2_{Q_\tau}, \tag{3.72}$$

discouraging thrust and azimuth commands larger than necessary to track the reference.

The last term penalizes $\mathbf{u}_v$, the rate of change of the thruster commands,

$$\big\|[\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R]^\top\big\|^2_R, \tag{3.73}$$

discouraging abrupt changes in thrust and azimuth between consecutive instants.

Summing (3.69)–(3.73) gives the stage cost

$$\begin{aligned}\ell(\mathbf{x}_v, \mathbf{u}_v) = \|\mathbf{p} - \mathbf{p}_{ref}\|^2_{Q_p} + q_\psi\big(1 - \cos(\psi - \psi_{ref})\big) + \|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|^2_{Q_\nu} \\ + \big\|[T_L, T_R, \alpha_L, \alpha_R]^\top\big\|^2_{Q_\tau} + \big\|[\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R]^\top\big\|^2_R.\end{aligned} \tag{3.74}$$

The terminal cost ℓ_f penalizes the last predicted state, so that the optimizer does not disregard where the vessel ends up at the end of the horizon. It equals the stage cost (3.74) without the input rate term, since no control input is chosen at the final node:

$$\ell_f(\mathbf{x}_v) = \|\mathbf{p} - \mathbf{p}_{ref}\|^2_{Q_p} + q_\psi\big(1 - \cos(\psi - \psi_{ref})\big) + \|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|^2_{Q_\nu} + \big\|[T_L, T_R, \alpha_L, \alpha_R]^\top\big\|^2_{Q_\tau}. \tag{3.75}$$

The thruster limits introduced in the predictive model enter the optimization problem as constraints.

Since thrust and azimuth are components of the state, their range limits define the admissible state set

$$\mathcal{X} = \left\{ \mathbf{x}_v : \begin{array}{ll} T_{L,min} \leq T_L \leq T_{L,max}, & T_{R,min} \leq T_R \leq T_{R,max}, \\ \alpha_{L,min} \leq \alpha_L \leq \alpha_{L,max}, & \alpha_{R,min} \leq \alpha_R \leq \alpha_{R,max} \end{array} \right\}, \tag{3.76}$$

while the limits on their rates of change, which act on the input, define the admissible input set

$$\mathcal{U} = \left\{ \mathbf{u}_v : \begin{array}{ll} |\dot{T}_L| \leq \dot{T}_{L,max}, & |\dot{T}_R| \leq \dot{T}_{R,max}, \\ |\dot{\alpha}_L| \leq \dot{\alpha}_{L,max}, & |\dot{\alpha}_R| \leq \dot{\alpha}_{R,max} \end{array} \right\}. \tag{3.77}$$

The last constraint couples the vessel to the UAV through the tether, and is the only one that depends on the other vehicle. The tether limits the distance between its attachment point on the vessel deck and the UAV to a maximum admissible separation d_{max}. The attachment point is vertically aligned with the origin of the vessel's body frame, at a fixed height z_t in the inertial frame, so its position is (x, y, z_t). Along the horizon, the USV controller uses the UAV's most recently computed predicted trajectory $\hat{\mathbf{p}}_a = [\hat{x}_a, \hat{y}_a, \hat{z}_a]^\top$, temporally resampled at the vessel's prediction nodes covered by it, to compute the squared distance between the attachment point and the UAV,

$$d^2 = (x - \hat{x}_a)^2 + (y - \hat{y}_a)^2 + (\hat{z}_a - z_t)^2. \tag{3.78}$$

The requirement $d^2 \leq d_{max}^2$ is enforced as a soft constraint through a slack variable $\sigma \geq 0$,

$$\underbrace{d^2 - d_{max}^2}_{h(\mathbf{x}_v)} \leq \sigma, \tag{3.79}$$

which is penalized by $\rho(\sigma)$ as in (3.56), with large weights μ_1 and μ_2 so that violations remain small whenever geometrically feasible.

Optimal Control Problem

At every sampling instant k, the kinematic and dynamic components of the initial state come from the vessel's current estimate, while the thruster components use the last command actually applied:

$$\mathbf{x}_{v,k} = \left[x_k, y_k, \psi_k, u_k, v_k, r_k,\ T_{L,k-1}, T_{R,k-1}, \alpha_{L,k-1}, \alpha_{R,k-1}\right]^\top. \tag{3.80}$$

Starting from the last applied command ensures that the rate limits in (3.61) are enforced relative to the actual thruster values, so the first input chosen is always attainable by the thrusters.

The references and the UAV's predicted trajectory are given data at each instant. Since the UAV's predicted trajectory can be shorter than the vessel's horizon, let $N_c \leq N$ denote the last node of the vessel's horizon covered by it. The tether constraint, and its slack, are enforced only for $i = 0, \ldots, N_c$.

Assembling the cost, the model, and the constraints gives the optimal control problem the USV solves

at every sampling instant:

$$\begin{aligned}
\min_{\hat{\mathbf{x}}_v,\,\hat{\mathbf{u}}_v,\,\hat{\sigma}} \quad & \sum_{i=0}^{N-1} \ell\big(\hat{\mathbf{x}}_{v,k+i|k}, \hat{\mathbf{u}}_{v,k+i|k}\big) + \ell_f\big(\hat{\mathbf{x}}_{v,k+N|k}\big) + \sum_{i=0}^{N_c} \rho\big(\hat{\sigma}_{k+i|k}\big) \\
\text{s.t.} \quad & \hat{\mathbf{x}}_{v,k|k} = \mathbf{x}_{v,k}, \\
& \hat{\mathbf{x}}_{v,k+i+1|k} = \boldsymbol{\Phi}_v\big(\hat{\mathbf{x}}_{v,k+i|k}, \hat{\mathbf{u}}_{v,k+i|k}\big), \quad i = 0, \dots, N-1, \\
& \hat{\mathbf{x}}_{v,k+i|k} \in \mathcal{X}, \quad i = 0, \dots, N, \\
& \hat{\mathbf{u}}_{v,k+i|k} \in \mathcal{U}, \quad i = 0, \dots, N-1, \\
& h\big(\hat{\mathbf{x}}_{v,k+i|k}\big) \le \hat{\sigma}_{k+i|k}, \quad \hat{\sigma}_{k+i|k} \ge 0, \quad i = 0, \dots, N_c.
\end{aligned} \tag{3.81}$$

The vessel's low-level actuators receive the integrated thruster commands $T_{L,k}, T_{R,k}, \alpha_{L,k}, \alpha_{R,k}$, obtained by integrating the first optimal input rate $\hat{\mathbf{u}}_{v,k|k}$ over the sampling interval T_s. These become the thruster components of the initial state at the subsequent sampling instant.

3.2.4 UAV NMPC

The UAV NMPC computes, at every sampling instant, the thrust and attitude commands that best track a reference trajectory over a finite prediction horizon, subject to the vehicle's dynamics, its actuation limits, and the tether workspace. It predicts the vehicle's motion with the model derived in Chapter 3.1, and solves a constrained optimization problem at each step to obtain the commands applied to the vehicle.

Predictive Model

The vehicle model defined in Chapter 3.1 assumed that the commanded attitude is tracked instantaneously by the low-level flight controller. In reality, the closed-loop attitude response of the autopilot exhibits a first-order lag characterized by a time constant $\tau > 0$:

$$\dot{\phi} = \frac{1}{\tau}\big(\phi_{\mathrm{cmd}} - \phi\big), \qquad \dot{\theta} = \frac{1}{\tau}\big(\theta_{\mathrm{cmd}} - \theta\big), \tag{3.82}$$

where ϕ, θ denote the actual Euler angles, $\phi_{\mathrm{cmd}}, \theta_{\mathrm{cmd}}$ are the attitude setpoints commanded to the autopilot, and τ is the attitude tracking time constant.

Furthermore, similarly to the thrusters of the USV, the attitude commands cannot change arbitrarily fast, and their rates of change are subject to physical slew-rate limits:

$$|\dot{\phi}_{\mathrm{cmd}}| \le \dot{\phi}_{max}, \quad |\dot{\theta}_{\mathrm{cmd}}| \le \dot{\theta}_{max}. \tag{3.83}$$

Imposing these limits directly as bounds in the optimization problem requires the rates of the commanded angles to be part of the control input vector,

$$\mathbf{u}_a = [\dot{\phi}_{\mathrm{cmd}}, \dot{\theta}_{\mathrm{cmd}}, T]^\top \in \mathbb{R}^3, \tag{3.84}$$

which in turn incorporates the commanded attitude into the state vector,

$$\mathbf{x}_a = [x, y, z, \dot{x}, \dot{y}, \dot{z},\ \phi, \theta,\ \phi_{\mathrm{cmd}}, \theta_{\mathrm{cmd}}]^\top \in \mathbb{R}^{10}. \tag{3.85}$$

With this state augmentation, range limits act on the state and rate limits act directly on the input. The vehicle heading ψ is aligned with the bearing to the reference target, as detailed below.

With $\mathbf{x}_a$ and $\mathbf{u}_a$ defined, the prediction model $\dot{\mathbf{x}}_a = \mathbf{f}_a(\mathbf{x}_a, \mathbf{u}_a)$ is assembled from kinematics, translational dynamics, attitude lag, and command integrators.

The kinematics are the trivial identity already used in (3.29),

$$\dot{x} = \dot{x}, \qquad \dot{y} = \dot{y}, \qquad \dot{z} = \dot{z}. \tag{3.86}$$

The translational dynamics follow (3.37), with the thrust term expanded, including the linear horizontal rotor drag opposing translational motion, and the tether force $\mathbf{F}^{\mathcal{I}}_{\mathrm{tether}} = [F^{\mathcal{I}}_{\mathrm{tether},x}, F^{\mathcal{I}}_{\mathrm{tether},y}, F^{\mathcal{I}}_{\mathrm{tether},z}]^\top$ given by (3.38)–(3.46):

$$\ddot{x} = \frac{T}{m}\big(\cos\psi \sin\theta \cos\phi + \sin\psi \sin\phi\big) - \frac{d_r}{m}\dot{x} + \frac{1}{m}F^{\mathcal{I}}_{\mathrm{tether},x}, \tag{3.87}$$

$$\ddot{y} = \frac{T}{m}\big(\sin\psi \sin\theta \cos\phi - \cos\psi \sin\phi\big) - \frac{d_r}{m}\dot{y} + \frac{1}{m}F^{\mathcal{I}}_{\mathrm{tether},y}, \tag{3.88}$$

$$\ddot{z} = \frac{T}{m}\cos\theta\cos\phi - g + \frac{1}{m}F^{\mathcal{I}}_{\mathrm{tether},z}. \tag{3.89}$$

The attitude dynamics combine the first-order tracking lag with the command rate integrators:

$$\dot{\phi} = \frac{1}{\tau}\big(\phi_{\mathrm{cmd}} - \phi\big), \qquad \dot{\theta} = \frac{1}{\tau}\big(\theta_{\mathrm{cmd}} - \theta\big), \tag{3.90}$$

$$\dot{\phi}_{\mathrm{cmd}} = \dot{\phi}_{\mathrm{cmd}}, \qquad \dot{\theta}_{\mathrm{cmd}} = \dot{\theta}_{\mathrm{cmd}}. \tag{3.91}$$

Discretization over a sampling interval T_s, following Section 3.2.2, gives the one-step map $\mathbf{\Phi}_a$ such that $\hat{\mathbf{x}}_{a,k+i+1|k} = \mathbf{\Phi}_a\big(\hat{\mathbf{x}}_{a,k+i|k}, \hat{\mathbf{u}}_{a,k+i|k}\big)$ approximates the solution of $\dot{\mathbf{x}}_a = \mathbf{f}_a(\mathbf{x}_a, \mathbf{u}_a)$ over one sampling interval.

Cost and Constraints

Let $\mathbf{p} = [x, y, z]^\top$ and $\mathbf{v} = [\dot{x}, \dot{y}, \dot{z}]^\top$ denote the position and velocity, with $\mathbf{p}_{ref}$ and $\mathbf{v}_{ref}$ the corresponding references.

Tracking the reference position and velocity is penalized quadratically as

$$\|\mathbf{p} - \mathbf{p}_{ref}\|^2_{Q_p} + \|\mathbf{v} - \mathbf{v}_{ref}\|^2_{Q_v}. \tag{3.92}$$

The next term penalizes ϕ, θ directly, rather than comparing them against any reference,

$$\big\|[\phi, \theta]^\top\big\|^2_{Q_{tilt}}, \tag{3.93}$$

discouraging attitude larger than necessary to track the reference.

The next term penalizes the yaw error, but rather than tracking a fixed yaw angle, ψ is aligned with the direction to a reference point $\mathbf{p}_{b,ref} = [x_{b,ref}, y_{b,ref}]^\top$, itself possibly varying along the horizon,

$$\mathbf{b} = \mathbf{p}_{b,ref} - \mathbf{p}_{xy}, \tag{3.94}$$

where $\mathbf{p}_{xy} = [x, y]^\top$. Since ψ is an angle, the periodicity issue discussed for (3.70) also applies here, and it is handled by expressing the error as an alignment between ψ and the bearing vector $\mathbf{b}$, rather than as a difference between two angles,

$$q_\psi \left(1 - \frac{\cos\psi\, b_x + \sin\psi\, b_y}{\sqrt{\|\mathbf{b}\|^2 + \epsilon}}\right), \tag{3.95}$$

where $q_\psi \geq 0$ is a scalar weight setting the relative importance of yaw tracking, and $\epsilon > 0$ is a small constant that keeps the term well defined when the UAV is above the reference point, where the bearing is undefined. This term is minimized when ψ points along $\mathbf{b}$.

The last term penalizes the input $\mathbf{u}_a$ relative to a nominal hover reference $\mathbf{u}_{a,ref} = [0, 0, mg]^\top$, since the thrust cannot be penalized toward zero without opposing gravity,

$$\|\mathbf{u}_a - \mathbf{u}_{a,ref}\|_R^2, \tag{3.96}$$

so that the attitude command rates are regularized toward zero, while the thrust is penalized only for deviating from the hover value.

Summing the tracking, tilt, yaw, and input penalties in (3.92), (3.93), (3.95), and (3.96) gives the stage cost

$$\begin{aligned}\ell(\mathbf{x}_a, \mathbf{u}_a) = \|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + \|\mathbf{v} - \mathbf{v}_{ref}\|_{Q_v}^2 + \left\|[\phi, \theta]^\top\right\|_{Q_{tilt}}^2 \\ + q_\psi \left(1 - \frac{\cos\psi\, b_x + \sin\psi\, b_y}{\|\mathbf{b}\|}\right) + \|\mathbf{u}_a - \mathbf{u}_{a,ref}\|_R^2.\end{aligned} \tag{3.97}$$

The terminal cost ℓ_f penalizes the last predicted state, so that the optimizer does not disregard where the UAV ends up at the end of the horizon. It equals the stage cost (3.97) without the input term, since no input is chosen at the final node:

$$\begin{aligned}\ell_f(\mathbf{x}_a) = \|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + \|\mathbf{v} - \mathbf{v}_{ref}\|_{Q_v}^2 + \left\|[\phi, \theta]^\top\right\|_{Q_{tilt}}^2 \\ + q_\psi \left(1 - \frac{\cos\psi\, b_x + \sin\psi\, b_y}{\|\mathbf{b}\|}\right).\end{aligned} \tag{3.98}$$

The vehicle's operating envelope bounds the altitude, translational velocity, and attitude. To guarantee safe clearance above the water surface and vessel deck, the vehicle is restricted to $z \geq z_{min}$. Furthermore, because horizontal and vertical velocities are limited by distinct physical factors, the vehicle's speed and tilt are bounded separately:

$$z \geq z_{min}, \quad \left\|[\dot{x}, \dot{y}]^\top\right\| \leq v_{xy,max}, \quad |\dot{z}| \leq v_{z,max}, \quad |\phi| \leq \phi_{max}, \quad |\theta| \leq \theta_{max}. \tag{3.99}$$

Unlike the thruster commands of the USV, the velocity and attitude are subject to external disturbances such as wind, which can push the true state past these bounds regardless of what the NMPC has commanded. The bounds of (3.99) are therefore softened as in (3.55), each with its own slack variable $\sigma_j \geq 0$, $j = 1, \ldots, 5$:

$$\underbrace{z_{min} - z}_{h_1(\mathbf{x}_a)} \leq \sigma_1, \tag{3.100}$$

$$\underbrace{\left\| [\dot{x}, \dot{y}]^\top \right\| - v_{xy,max}}_{h_2(\mathbf{x}_a)} \leq \sigma_2, \tag{3.101}$$

$$\underbrace{|\dot{z}| - v_{z,max}}_{h_3(\mathbf{x}_a)} \leq \sigma_3, \tag{3.102}$$

$$\underbrace{|\phi| - \phi_{max}}_{h_4(\mathbf{x}_a)} \leq \sigma_4, \tag{3.103}$$

$$\underbrace{|\theta| - \theta_{max}}_{h_5(\mathbf{x}_a)} \leq \sigma_5. \tag{3.104}$$

Each slack is penalized by $\rho_j(\sigma_j)$ as in (3.56), with weights $\mu_{1,j}$ and $\mu_{2,j}$.

The attitude command rate limits of the predictive model (3.83) and the thrust range are the only hard constraints, and they act on the input, defining the admissible input set

$$\mathcal{U} = \left\{ \mathbf{u}_a : \begin{array}{l} |\dot{\phi}_{\mathrm{cmd}}| \leq \dot{\phi}_{max}, \quad |\dot{\theta}_{\mathrm{cmd}}| \leq \dot{\theta}_{max}, \\ T_{min} \leq T \leq T_{max} \end{array} \right\}. \tag{3.105}$$

The remaining constraints couple the UAV to the vessel through the tether, and are the only ones that depend on the other vehicle. First, as for the vessel, the tether limits the distance between the UAV and the tether attachment point, described in Section 3.2.3, to the maximum admissible separation d_{max}. The attachment point on the USV is located at $(\hat{x}_v, \hat{y}_v, z_t)$, where the horizontal position $\hat{\mathbf{p}}_v = [\hat{x}_v, \hat{y}_v]^\top$ at each node is taken from the vessel's most recently computed predicted trajectory, temporally resampled at the UAV's prediction nodes, and z_t is the height of the attachment point. Since the vessel's horizon covers that of the UAV, the constraint is enforced at every node. Denoting by $\mathbf{r} = [x - \hat{x}_v, y - \hat{y}_v, z - z_t]^\top$ the relative position vector between the vehicle and the attachment point, the squared separation distance is

$$d^2 = \|\mathbf{r}\|^2 = (x - \hat{x}_v)^2 + (y - \hat{y}_v)^2 + (z - z_t)^2. \tag{3.106}$$

The requirement $d^2 \leq d_{max}^2$ is enforced as a soft constraint through a slack variable $\sigma_6 \geq 0$,

$$\underbrace{d^2 - d_{max}^2}_{h_6(\mathbf{x}_a)} \leq \sigma_6, \tag{3.107}$$

which is penalized by $\rho_6(\sigma_6)$ as in (3.56), with large weights $\mu_{1,6}$ and $\mu_{2,6}$ so that violations remain small.

Similarly, to prevent tension spikes during tether deployment, the rate at which the inter-vehicle distance increases must not exceed the maximum pay-out rate of the winch mechanism, $\dot{d} \leq \dot{d}_{max}$. Differ-

entiating $d = \|\mathbf{r}\|$ with respect to time yields the projection of the relative velocity $\dot{\mathbf{r}} = [\dot{x} - \hat{\dot{x}}_v,\ \dot{y} - \hat{\dot{y}}_v,\ \dot{z}]^\top$ onto the tether line of sight:

$$\dot{d} = \frac{\mathrm{d}}{\mathrm{d}t}\|\mathbf{r}\| = \frac{\mathbf{r}^\top \dot{\mathbf{r}}}{d}. \tag{3.108}$$

This formulation restricts only radial motion that moves away from the vessel ($\dot{d} > 0$), leaving tangential and approaching motions unconstrained. Enforcing $\dot{d} \leq \dot{d}_{max}$ with a small regularization $\epsilon > 0$ to avoid numerical singularity when $d \to 0$ gives the softened constraint:

$$\underbrace{\frac{\mathbf{r}^\top \dot{\mathbf{r}}}{\sqrt{d^2 + \epsilon}} - \dot{d}_{max}}_{h_7(\mathbf{x}_a)} \leq \sigma_7, \tag{3.109}$$

which is penalized by $\rho_7(\sigma_7)$ as in (3.56).

Together, the minimum altitude z_{min} and the maximum tether separation d_{max} delimit the workspace of the UAV, illustrated in Figure 3.6.

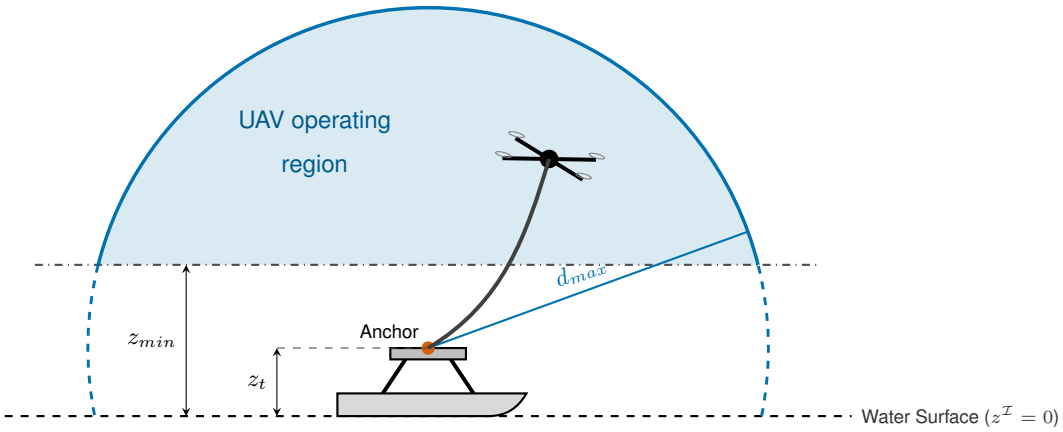


Figure 3.6: Operating region of the UAV.

All seven state constraints thus have the form $h_j(\mathbf{x}_a) \leq \sigma_j$, and since every one of them is softened, there are no hard constraints on the state. With the slacks stacked as $\boldsymbol{\sigma} = [\sigma_1, \ldots, \sigma_7]^\top$, the combined penalty is

$$P(\boldsymbol{\sigma}) = \sum_{j=1}^{7} \rho_j(\sigma_j). \tag{3.110}$$

Optimal Control Problem

At every sampling instant k, the translational and physical attitude components of the initial state come from the vehicle's current state estimate, while the commanded attitude components use the last commands actually applied:

$$\mathbf{x}_{a,k} = \left[x_k, y_k, z_k, \dot{x}_k, \dot{y}_k, \dot{z}_k,\ \phi_k, \theta_k,\ \phi_{\mathrm{cmd},k-1}, \theta_{\mathrm{cmd},k-1}\right]^\top. \tag{3.111}$$

The references and the vessel's most recently computed predicted trajectory are given data at each

instant.

Assembling the cost, the model, and the constraints gives the optimal control problem the UAV solves at every sampling instant, with the slacks $\hat{\sigma}_{j,k+i|k}$ entering as additional decision variables:

$$
\begin{aligned}
\min_{\hat{\mathbf{x}}_a,\,\hat{\mathbf{u}}_a,\,\hat{\boldsymbol{\sigma}}} \quad & \sum_{i=0}^{N-1} \ell\big(\hat{\mathbf{x}}_{a,k+i|k}, \hat{\mathbf{u}}_{a,k+i|k}\big) + \ell_f\big(\hat{\mathbf{x}}_{a,k+N|k}\big) + \sum_{i=0}^{N} P\big(\hat{\boldsymbol{\sigma}}_{k+i|k}\big) \\
\text{s.t.} \quad & \hat{\mathbf{x}}_{a,k|k} = \mathbf{x}_{a,k}, \\
& \hat{\mathbf{x}}_{a,k+i+1|k} = \boldsymbol{\Phi}_a\big(\hat{\mathbf{x}}_{a,k+i|k}, \hat{\mathbf{u}}_{a,k+i|k}\big), \quad i = 0, \dots, N-1, \\
& \hat{\mathbf{u}}_{a,k+i|k} \in \mathcal{U}, \quad i = 0, \dots, N-1, \\
& h_j\big(\hat{\mathbf{x}}_{a,k+i|k}\big) \le \hat{\sigma}_{j,k+i|k}, \quad \hat{\sigma}_{j,k+i|k} \ge 0, \quad j = 1, \dots, 7, \quad i = 0, \dots, N.
\end{aligned}
\tag{3.112}
$$

The vehicle's low-level autopilot receives the attitude setpoints $\phi_{\mathrm{cmd},k}, \theta_{\mathrm{cmd},k}$, obtained by integrating the first optimal rates $\hat{\dot{\phi}}_{\mathrm{cmd},k|k}, \hat{\dot{\theta}}_{\mathrm{cmd},k|k}$ over T_s, together with the thrust command $\hat{T}_{k|k}$ of the first optimal input $\hat{\mathbf{u}}_{a,k|k}$. These become the commanded attitude components of the initial state at the subsequent sampling instant.

Chapter 4

Implementation

Chapter 3 defines the models and the controllers of the tethered USV–UAV system, but their behaviour and performance can only be assessed once they are implemented and executed. This depends on how the controllers interact with the vehicles, the tether, and the marine environment, which cannot be studied analytically.

For this reason, a realistic simulation environment, integrated with Robot Operating System (ROS) 2, was developed to support the design of the controllers. It represents the vehicles and the tether, and provides controlled and repeatable conditions under which the controllers are tested. The controllers were then implemented to run in real time and adapted to the vehicles used in simulation. Finally, manoeuvres and missions were defined to evaluate the performance of the controllers in representative operations.

4.1 Simulation Environment

Simulation is the natural first step in the development of the controllers, since it allows them to be tested and adjusted before being used on the real vehicles. The simulations in this work are run in Gazebo [23], a robotics simulator. A simulator is a program that creates a virtual environment, places robots in it, and calculates how they would behave in reality. At each step, it receives the commands sent to the robots, computes the forces acting on them, updates their positions accordingly, and returns the measurements that their sensors would produce. To the controller, the simulated robot behaves like a real one.

This allows controllers to be developed through many fast iterations, and different scenarios to be repeated under the same conditions, which makes the comparison between controllers and parameter choices fair. Sensor noise and estimation error can also be included in simulation, but they can be left out during development, so that the behaviour of the controllers is analysed on its own.

4.1.1 World

The space where the simulation takes place is called the world. It contains the environment around the robots, such as the terrain, obstacles, and light, and defines the rules that apply inside it, such as the way time advances, the physical effects that are calculated, and the sensors that can be simulated. The robots are not part of the world. They are added to it afterwards, and what happens to them results from the interaction between their own models and the world in which they are placed.

A world is defined in a text file that lists its elements and the plugins that enable its behaviour. Plugins are software modules that add functionality to the simulator, such as the calculation of buoyancy or the simulation of sensors.

The world used in this work represents an open water area, shown in Figure 4.1. It consists of a sea surface and a seabed below it, and the buoyancy is calculated from the density of the water, which allows the hull of the USV to float. The world also enables the simulation of the sensors that the vehicles can use to estimate their state, such as inertial, magnetic, pressure, and satellite positioning sensors.

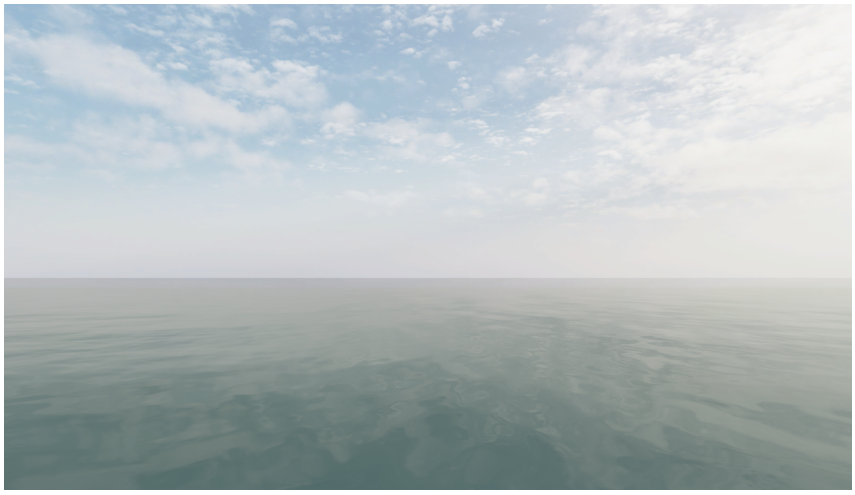


Figure 4.1: View of the simulated world

The vehicles operate at sea, where wind, waves, and currents act on them. The world allows these disturbances to be included, so that the controllers can be tested under conditions that their models do not predict. Each disturbance acts through a different physical path. Wind applies a force on the bodies exposed to the air, waves change the height of the water surface and therefore the buoyancy of the hull, and currents move the water itself, which changes the drag on the submerged bodies. These disturbances can be enabled, disabled, or adjusted for each test.

4.1.2 Wave Adaptive Modular Vessel

The Wave Adaptive Modular Vessel (WAM-V) is a catamaran with two inflatable hulls connected by a central payload deck, used in maritime robotics competitions. In this work it is the USV of the tethered system. It was chosen mainly because its payload deck is large enough to serve as a platform for the takeoff and landing of the UAV. In addition, a complete and publicly available model of the vessel exists

in the Virtual RobotX (VRX) simulation framework [24], so that the work could focus on the control and not on building and validating a vessel model. Figure 4.2 shows the vehicle in the simulated world.

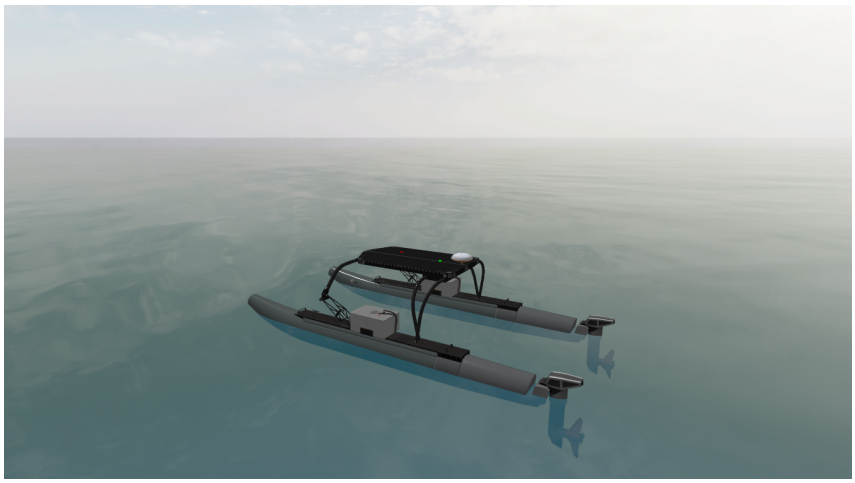
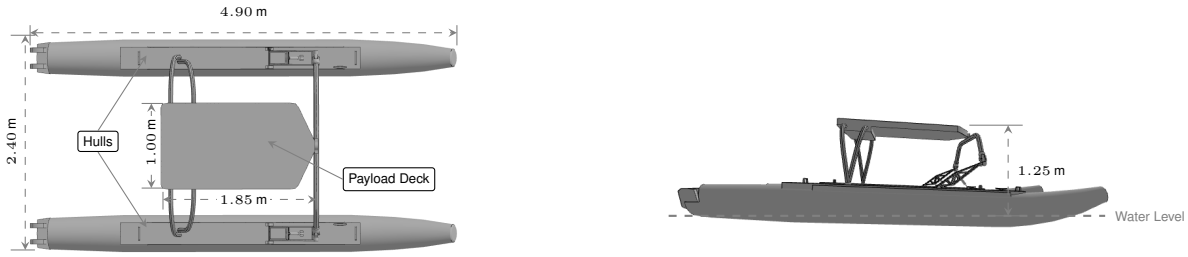


Figure 4.2: The WAM-V catamaran in the simulated world.

In the simulator, a vehicle is described by a model that lists its bodies, the joints that connect them, and for each body its mass, its inertia, and its geometry. The WAM-V is represented by a main rigid body, which includes the hulls, the deck, and the batteries, with a mass of $227\,\mathrm{kg}$ and moments of inertia of 181, 408, and $495\,\mathrm{kg}\cdot\mathrm{m}^2$ about the longitudinal, lateral, and vertical axes. The two thrusters and the sensors are separate bodies attached to it through joints. The thrusters can rotate to steer the vessel, while the joints of the sensors are fixed. Including all bodies, the total mass is $259.10\,\mathrm{kg}$.

The catamaran has an overall length of $4.90\,\mathrm{m}$ and a beam of $2.40\,\mathrm{m}$. The payload deck, with $1.85\,\mathrm{m}$ in length and $1.00\,\mathrm{m}$ in width, is supported by the two hulls at about $1.25\,\mathrm{m}$ above the water surface, as shown in Figure 4.3.



(a) Top view.

(b) Side view.

Figure 4.3: WAM-V catamaran layout.

The behaviour of the vessel on the water is determined by how the simulator reproduces the forces that the water exerts on it. The hulls are kept afloat by a buoyancy force, which is calculated from the volume of each hull below the water surface. When waves are present, this volume changes along the hulls and the catamaran pitches and rolls accordingly. As the vessel moves, the water also resists its motion through drag forces and moments applied in the six degree of freedoms (DOFs). For a velocity

ν_i along or about each degree of freedom, the drag is

$$D_i = -\left(d_i^{\text{lin}} + d_i^{\text{quad}}\,|\nu_i|\right)\nu_i, \tag{4.1}$$

where d_i^{lin} and d_i^{quad} are the linear and quadratic coefficients. The negative sign makes the drag oppose the motion, and the linear term dominates at low speeds while the quadratic term dominates at higher speeds. This makes the vehicle slow down naturally when the propulsion is cut off. The coefficients of each degree of freedom, given in Table 4.1, are those provided by the VRX model.

Table 4.1: Hydrodynamic damping coefficients of the simulated WAM-V.

Damping Mode	Linear Coefficient	Quadratic Coefficient	Unit
Surge	100.00	150.00	N$/$(m/s) or N$/$(m/s)^2
Sway	100.00	100.00	N$/$(m/s) or N$/$(m/s)^2
Heave	500.00	–	N$/$(m/s)
Roll	300.00	600.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2
Pitch	900.00	900.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2
Yaw	800.00	800.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2

The vehicle is propelled by two azimuth thrusters, one at the stern of each hull, and it steers by rotating them instead of using a rudder. To move, the model needs to receive, for each thruster, the thrust force that it must produce, in newtons, and its steering angle. The angle is limited to ± 30 degrees, to avoid twisting the power cables, and its rate of change to 10 degrees per second. The thrust is limited to 510.0 N in the forward direction, a value representative of the electric outboard motors used on the WAM-V in robotics competitions, such as the Torqeedo Cruise 2.0 [25]. In reverse, the propeller is less efficient, which is modelled with a multiplier of 0.746 that limits the thrust to -380.0 N.

4.1.3 X500 Quadrotor

The Holybro X500 v2 [26] is a quadrotor, and in this work it is the UAV of the tethered system. It was chosen mainly because it is heavier and more stable than small multirotors, which makes it better suited to fly with a tether attached to it. Its size is also compatible with the payload deck of the WAM-V, leaving enough room for takeoff and landing, and a complete model of the vehicle is available in the simulation ecosystem of the PX4 autopilot [27], so that the work could focus on the control and not on building and validating a vehicle model. Figure 4.4 shows the vehicle in the simulated world.

The x500 is described in the simulator in the same way as the WAM-V, by a model that lists its bodies, the joints that connect them, and the mass and inertia of each one. It consists of a main rigid body, which includes the frame and the sensors, and of four rotors, which are separate bodies connected to it through rotating joints. Their masses, inertias, and positions are given in Table 4.2. The rotors are 0.348 m apart along the sides of the frame and about 0.49 m apart along the diagonal. The landing gear is 0.264 m wide, and the motor mounts are 0.26 m above the ground, as shown in Figure 4.5.

The vehicle is lifted by the thrust that its rotors produce by pushing air downwards. The model receives, for each rotor i, a command with its angular speed ω_i, in rad/s, and each rotor produces a

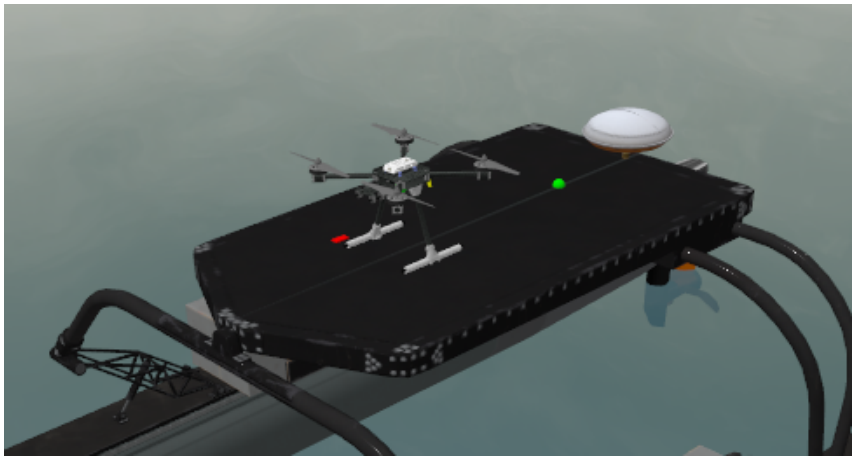
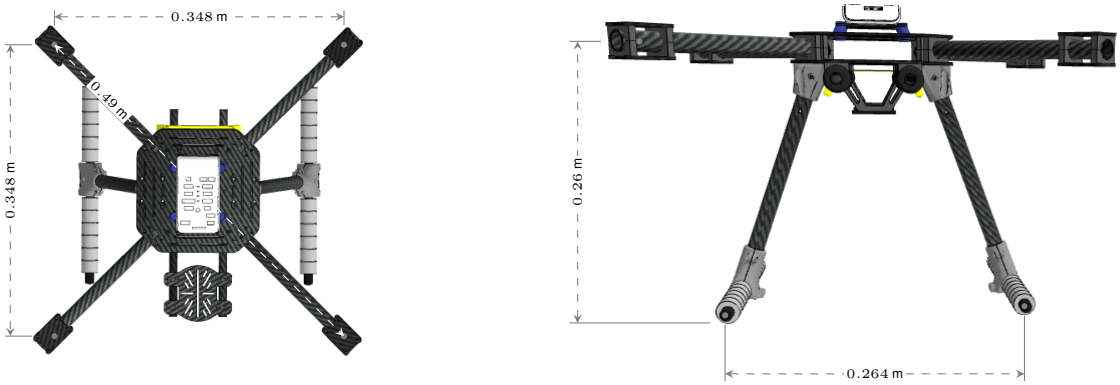


Figure 4.4: The x500 quadrotor in the simulated world.



(a) Top view.

(b) Front view.

Figure 4.5: x500 quadrotor layout.

Table 4.2: Parameters of the simulated x500, based on the PX4 model [28].

Parameter	**Value**	**Unit**
Main body mass	2.0	kg
Rotor mass (each)	0.016	kg
Total mass	2.06	kg
Moment of inertia, longitudinal and lateral axes	0.0217	kg · m^2
Moment of inertia, vertical axis	0.040	kg · m^2
Rotor position, along each horizontal axis	$\pm$0.174	m
Thrust coefficient k_F	1.11×10^{-5}	N/(rad/s)^2
Moment coefficient k_M	0.016	m
Maximum rotor speed	1000	rad/s
Rotor time constant, accelerating	12.5	ms
Rotor time constant, decelerating	25	ms

thrust force F_i and a reaction moment M_i that depend on it [29],

$$F_i = k_F\,\omega_i^2, \qquad M_i = s_i\,k_M\,F_i, \tag{4.2}$$

where k_F is the thrust coefficient, k_M is the moment coefficient, and $s_i = \pm 1$ gives the sign of the moment, which depends on the direction of rotation. Rotors on the same diagonal rotate in the same direction and adjacent rotors in opposite directions, so that the reaction moments balance each other when the vehicle hovers. The model also includes the aerodynamic drag of the rotors.

The rotors have limits that the model reproduces. The speed of each rotor does not change instantly, and it follows the command with the time constants given in Table 4.2, up to the maximum speed given in the same table. At this speed, each rotor produces about 11.1 N, and the four together produce 44.5 N. As the weight of the vehicle is 20.2 N, the thrust-to-weight ratio is about 2.2. In hovering, the rotors produce about 45% of their maximum thrust, at a speed of about 675 rad/s, which leaves about 24 N available to manoeuvre and to counter the force of the tether.

4.1.4 Tether

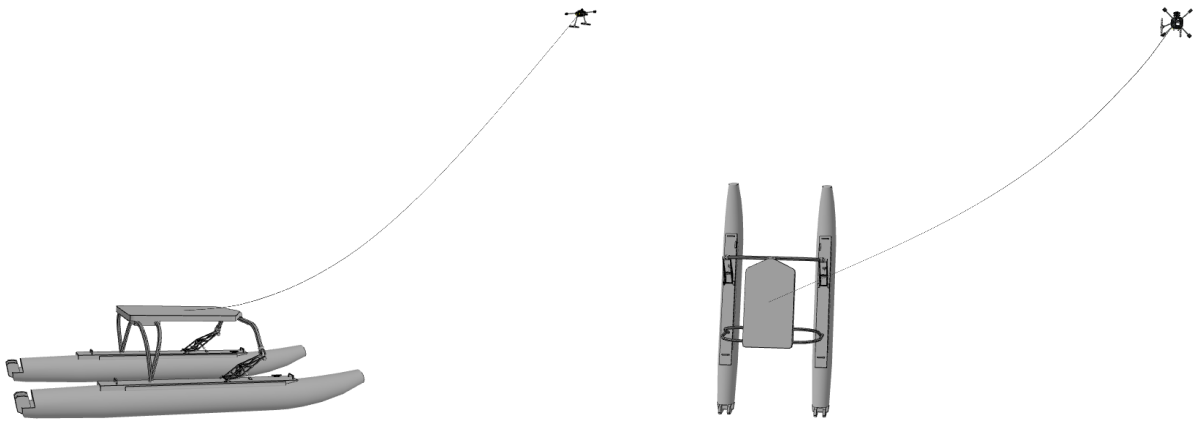
The tether couples the two vehicles, so the behaviour of the system depends on how it is represented. To evaluate the controllers as close as possible to real operation, they should be tested against a tether model that is more detailed than the one used in their design. The model used by the controllers, described in Section 3.1, approximates the tether by a catenary linearised around a constant slack. This simplification keeps the computational cost low, since the controllers evaluate the model at every step of the prediction horizon within each sampling interval. The simulated tether does not have this constraint, so a more detailed model can be used. Since this type of cable is not included in the vehicle models, it is simulated separately and coupled to them.

The tether is simulated with MoorDyn [30], a tool that represents a cable as a chain of lumped masses connected by segments that behave as springs and dampers. Each segment resists stretching and bending, while the weight and the drag of the cable are calculated at the nodes between segments. The slack and the tension of the cable result from the motion of these nodes, without assuming a shape for the cable. MoorDyn runs separately from Gazebo, which does not represent the cable itself but receives its effect on the vehicles through the forces that MoorDyn calculates. The cable is therefore displayed in a separate visualisation tool, Foxglove [31], as shown in Figure 4.6.

In this type of model, the mass of the cable is concentrated at the nodes, and the segments between them have no mass, as shown in Figure 4.7a. The motion of each free node i follows from Newton's second law, with the forces shown in Figure 4.7b,

$$m_i\,\ddot{\mathbf{p}}_i = \mathbf{T}_{i-1} + \mathbf{T}_i + \mathbf{F}_i^{\text{bend}} + m_i\mathbf{g} + \mathbf{F}_i^{\text{drag}}, \tag{4.3}$$

where m_i and $\mathbf{p}_i$ are the mass and the position of the node, $\mathbf{T}_{i-1}$ and $\mathbf{T}_i$ are the tensions of the two adjacent segments, $\mathbf{F}_i^{\text{bend}}$ is the force due to the bending of the cable at the node, and $\mathbf{F}_i^{\text{drag}}$ is the drag,



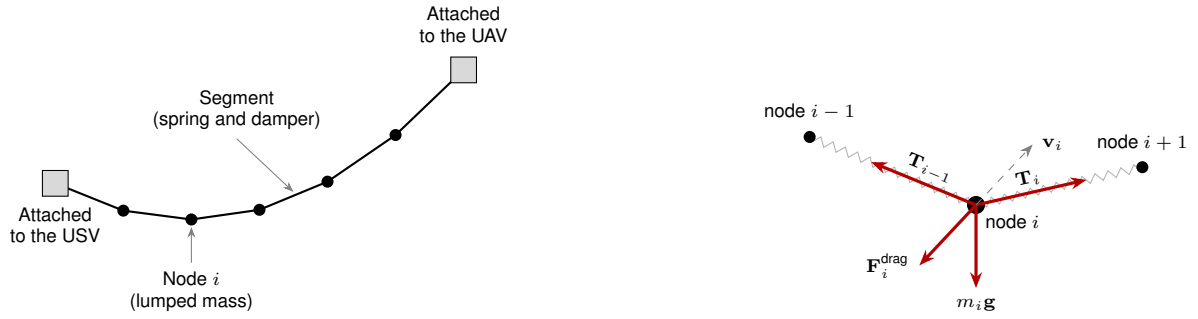
(a) Side view. (b) Top view.

Figure 4.6: The tether between the USV and the UAV, displayed in Foxglove.

which opposes the velocity of the node. The magnitude of the tension of a segment depends on its elongation and on the rate at which it changes,

$$T = \begin{cases} EA\,\eta + BA\,\dot{\eta}, & \eta > 0, \\ 0, & \eta \leq 0, \end{cases} \qquad \eta = \frac{\ell - \ell_0}{\ell_0}, \tag{4.4}$$

where ℓ and ℓ_0 are the current and the unstretched lengths of the segment, EA is the axial stiffness, and BA is the damping coefficient, which is defined through the damping ratio given in Table 4.3. A segment that is shorter than its unstretched length carries no tension, so the cable can go slack. The accelerations are then integrated in time to obtain the new position and velocity of every node. The end nodes are not integrated, since their motion is imposed by the vehicles, and the forces at these nodes are the forces that the cable exerts on them.



(a) Cable divided into nodes and segments. (b) Forces on a node.

Figure 4.7: Lumped-mass model of the tether. Bending forces are not shown.

The parameters of the model are the mass per unit length, which gives the mass m_i of each node in (4.3), the axial stiffness EA and the damping ratio, which define the tension in (4.4), the bending stiffness, and the drag coefficients. Their values are given in Table 4.3, together with the number of segments and the time step used to integrate the equations. The mass per unit length and the diameter are based on commercial micro-tethers for drones that provide power and data [32, 33], and the axial

stiffness is estimated from the tensile strength of these tethers. The length of the cable is not fixed, and it is changed by the winch described in Section 4.1.5.

Table 4.3: Parameters of the simulated tether.

Parameter	**Value**	**Unit**
Diameter	3	mm
Mass per unit length	0.020	kg/m
Axial stiffness, EA	5×10^4	N
Damping ratio (relative to critical)	0.8	–
Bending stiffness	2×10^{-5}	$\mathrm{N \cdot m^2}$
Transverse drag coefficient	1.2	–
Axial drag coefficient	0.005	–
Number of segments	30	–
Integration time step	0.02	ms

The tether is coupled to the vehicles by a ROS node, as shown in Figure 4.8. At each step, the node reads the position and the velocity of the two attachment points, which are the tether anchor on the deck of the USV and the centre of the body of the UAV, and gives them to MoorDyn. MoorDyn imposes them as the motion of the ends of the cable, integrates the cable over the step, and returns the force vector at each attachment point, which is applied to the vehicles in Gazebo.

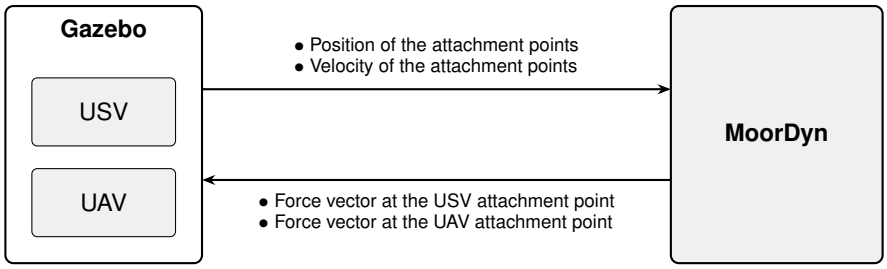


Figure 4.8: Data exchange between the simulation and the tether.

4.1.5 Tether Winch

The model of the tether used by the controllers in Section 3.1 assumes that its slack is kept close to a constant value ε_0, which requires a mechanism that adjusts the length of the cable as the vehicles move. In a real system, this can be achieved by a winch system on the vessel, which spools the cable in and out. In simulation, this mechanism is represented by a virtual winch that varies the length of the cable.

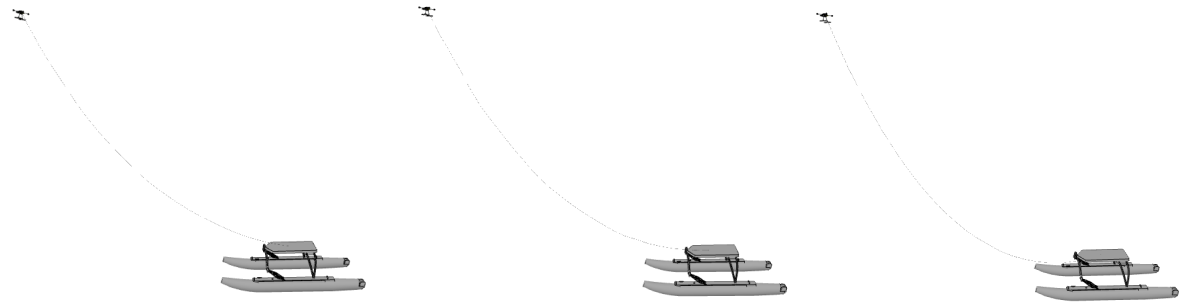
At each step, the virtual winch reads the distance d between the attachment points and computes the length that gives the desired slack,

$$L_{\text{ref}} = \frac{d}{1 - \varepsilon_0}, \tag{4.5}$$

which it then applies as the unstretched length of the cable in MoorDyn. A real system cannot follow the variations of the distance instantly, because of the speed of the winch and the delay in the measurement of the distance. To account for this limitation, the rate of change of the length is limited to v_{winch}, so that

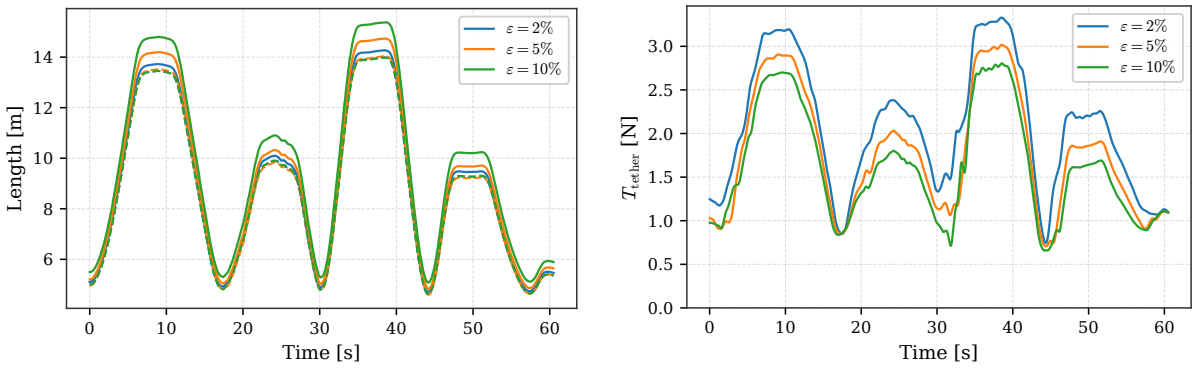
the slack deviates from ε_0 when the distance changes faster than the system can follow. The values used are $\varepsilon_0 = 0.05$ and $v_{\text{winch}} = 5$ m/s.

To evaluate the effect of the slack ratio, the same trajectory of the UAV was flown with $\varepsilon_0 = 2\%$, 5%, and 10%. A larger slack makes the cable sag more between the vehicles (Figure 4.9), and the length of the cable follows the distance between them, staying above it by a margin that grows with ε_0 (Figure 4.10a). The tension at the UAV follows the distance, since the weight of the cable that it supports increases with it, and decreases as the slack increases (Figure 4.10b). Table 4.4 quantifies these results. The maximum distance between the vehicles is about 14 m in the three runs, differing by 5 cm at most, which confirms that the trajectory is the same. From $\varepsilon_0 = 2\%$ to 10%, the mean tension decreases from 2.07 N to 1.62 N, a reduction of 22%, and the maximum from 3.38 N to 2.88 N.



(a) $\varepsilon_0 = 2\%$. (b) $\varepsilon_0 = 5\%$. (c) $\varepsilon_0 = 10\%$.

Figure 4.9: Shape of the tether for different slack ratios, displayed in Foxglove.



(a) Distance (dashed) and cable length (solid). (b) Tension at the UAV.

Figure 4.10: Distance between the attachment points, length of the cable, and tension at the UAV for the same trajectory and different slack ratios.

Table 4.4: Tension at the UAV and maximum distance between the vehicles for different slack ratios, over the 60 s of the trajectory.

ε_0	**Mean** (N)	**Maximum** (N)	**Std. dev.** (N)	**Max. distance** (m)
2%	2.07	3.38	0.73	13.99
5%	1.78	3.08	0.69	14.03
10%	1.62	2.88	0.64	13.98

These tensions are small compared with the weight of the UAV, which is 20.2 N, so the value of ε_0 is

not critical for the system. A value of 5% is therefore adopted, as a compromise between the amount of loose cable, which grows with ε_0, and the tension, which decreases with it.

4.2 Model Validation

Before incorporating the vehicle and tether models into the real-time controllers developed in Chapter 3, their accuracy and validity must be verified against the high-fidelity simulation environment described in Section 4.1. This ensures that the simplified analytical formulations adequately capture the dominant physical dynamics while remaining computationally tractable for online execution.

4.2.1 USV Dynamics Validation

The three-degree-of-freedom planar dynamic model derived in Section 3.1.2 and the thruster allocation described in Section 3.1.2 provide the predictive backbone for the USV trajectory tracking controllers. This control-oriented model intentionally simplifies the physical system by neglecting out-of-plane heave, pitch, and roll motions, as well as wave-induced radiation and diffraction forces. Before embedding this formulation into the real-time optimization loop of the non-linear predictive controller, its ability to reproduce the vessel's planar motions must be verified against the full six-degree-of-freedom physics of the Gazebo simulator.

Rather than relying on empirical grey-box fitting or tuning arbitrary parameters against experimental data, the validation is conducted strictly using the first-principles nominal parameters of the simulation environment. The total vessel mass is set to $m = 259.10\,\text{kg}$, comprising the $227.0\,\text{kg}$ catamaran hull, the two $15.0\,\text{kg}$ stern thruster assemblies, and $2.1\,\text{kg}$ of sensors and propeller pods. The yaw moment of inertia about the centre of gravity is $I_z = 695.90\,\text{kg}\cdot\text{m}^2$, obtained by combining the base hull inertia of $495.0\,\text{kg}\cdot\text{m}^2$ with the parallel axis theorem contribution of the two outboard thrusters mounted at longitudinal distance $x_{\text{arm}} = 2.3738\,\text{m}$ and lateral distance $y_{\text{arm}} = 1.0271\,\text{m}$. The hydrodynamic drag parameters are adopted directly from the simulator configuration given in Table 4.1, with linear and quadratic surge damping of $X_u = 100.0\,\text{N}/(\text{m/s})$ and $X_{uu} = 150.0\,\text{N}/(\text{m/s})^2$, sway damping of $Y_v = 100.0\,\text{N}/(\text{m/s})$ and $Y_{vv} = 100.0\,\text{N}/(\text{m/s})^2$, and yaw damping of $N_r = 800.0\,\text{N}\cdot\text{m}/(\text{rad/s})$ and $N_{rr} = 800.0\,\text{N}\cdot\text{m}/(\text{rad/s})^2$.

To excite the full operating envelope of the vessel, a dynamic manual joystick maneuver was recorded in the simulator over $63.84\,\text{s}$, covering a total traversed distance of $61.29\,\text{m}$. The telemetry records the physical actuator commands dispatched to the port and starboard azimuth thrusters, shown in Figure 4.11. The commanded thrust spans the full bidirectional range, reaching the forward saturation limit of $+510.0\,\text{N}$ and the reverse limit of $-380.0\,\text{N}$, while the azimuth angles rotate dynamically up to $\pm 24^\circ$.

The nominal model was integrated purely open-loop using a fourth-order Runge-Kutta scheme at $50\,\text{Hz}$, driven exclusively by the actuator commands mapped through equations (3.23) and (3.24). Figure 4.12 compares the predicted body-fixed velocities against the ground-truth telemetry from the simulator. The surge velocity u follows the sharp accelerations, cruising phases, and aggressive reverse

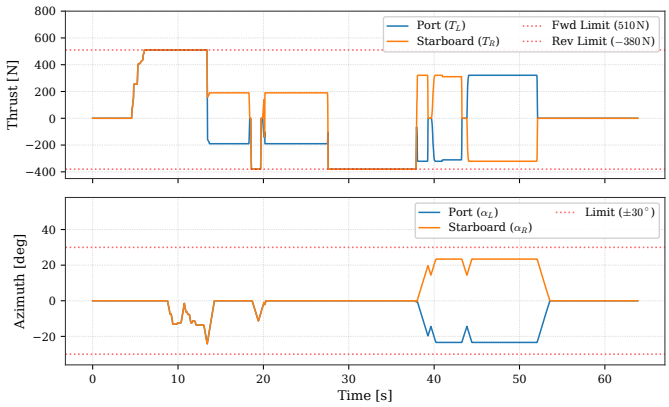


Figure 4.11: USV thruster inputs during the validation maneuver.

braking transitions with high fidelity, achieving a coefficient of determination of $R^2 = 99.90\%$ and an root-mean-square error (RMSE) of 0.0350 m/s. The sway velocity v, induced by the lateral components of the angled thrusters, is captured with $R^2 = 99.77\%$ and an RMSE of 0.0284 m/s. The yaw rate r reproduces both transient steering peaks and sustained rotations, yielding $R^2 = 98.94\%$ and an RMSE of $0.9925°/s$.

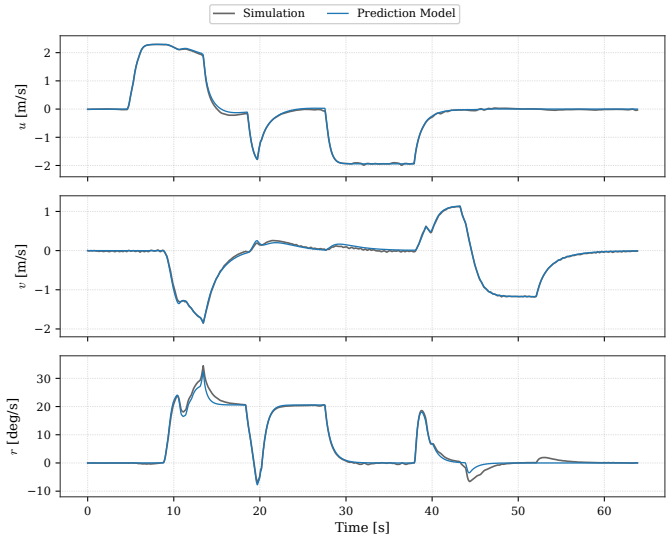


Figure 4.12: USV body-fixed velocities validation.

The kinematic validity of the planar model over extended horizons is demonstrated in Figure 4.13. Over the entire 63.84 s run, the predicted vessel heading ψ remains tightly aligned with the simulated catamaran, obtaining an R^2 of 99.78% and an RMSE of $7.0173°$. In the horizontal plane, the cumulative Euclidean position error between the open-loop integration and the ground truth reaches an RMSE of 2.190 m, which corresponds to only 3.57% of the total distance traveled over 61.29 m.

When evaluated in the operational context of the non-linear model predictive controller over the receding horizon $T_h = 2.0$ s ($N = 100$ integration steps at 50 Hz), the local predictive capability of the first-principles formulation is exceptionally high. As shown in Figure 4.13, the predicted trajectory segments track the actual vessel displacements with sub-decimetre precision. Figure 4.14 illustrates the error growth as a function of the prediction horizon: at the full 2.0 s horizon, the position error achieves an RMSE of only 0.0411 m (4.1 cm) with a 95th percentile of 0.0793 m (7.9 cm), and the velocity error

reaches an RMSE of $0.0402\,\mathrm{m/s}$.

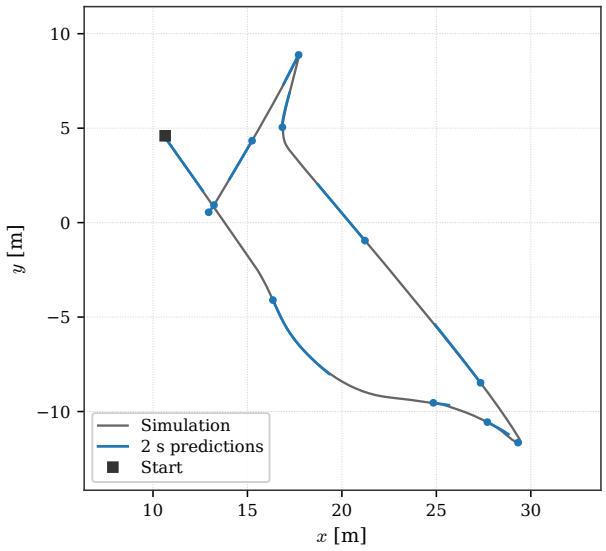
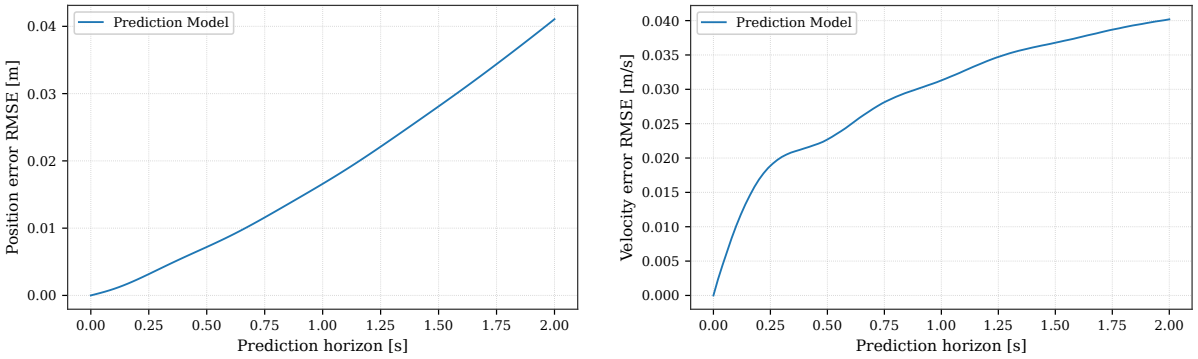


Figure 4.13: Two-second open-loop predictions along the USV trajectory.



(a) Position error RMSE.

(b) Velocity error RMSE.

Figure 4.14: USV open-loop prediction error RMSE as a function of the prediction horizon.

Table 4.5 provides a quantitative summary of the validation metrics across all state variables, covering both the full-run open-loop simulation and the receding-horizon predictions. The close agreement across all velocity and kinematic states confirms that the first-principles 3-DOF formulation and its thruster allocation equations capture the vessel dynamics with sufficient accuracy for model-based predictive control, without requiring grey-box parameter estimation.

4.2.2 UAV Dynamics Validation

What must the model capture, and why? The translational model of Section 3.1.3, with the closed-form tether force of Section 3.1.3, is the predictive model of the UAV NMPC of Section 3.2.4. For the controller, what matters is that it predicts the position and the velocity of the vehicle over the horizon $NT_s = 1\,\mathrm{s}$ from the attitude and the thrust that are commanded. This relies on three assumptions: the vehicle reaches the attitude that is commanded, the commanded thrust is the force that the rotors produce, and a point mass subject to thrust, gravity, and tether is a sufficient description of the motion.

Table 4.5: USV model validation metrics: open-loop full run (63.8 s) and receding horizon ($T_h = 2.0$ s).

State	Unit	RMSE	MAE	R^2 (%)	Max Error
Continuous Open-Loop Run (63.84 s)					
Surge velocity u	m/s	0.0350	0.0235	99.90	0.1372
Sway velocity v	m/s	0.0284	0.0217	99.77	0.1010
Yaw rate r	deg/s	0.9925	0.5989	98.94	3.5991
Heading ψ	deg	7.0173	5.8392	99.78	11.0267
Longitudinal position x	m	1.9996	1.4632	89.65	3.2867
Lateral position y	m	0.8926	0.6802	98.17	1.3779
Euclidean position e_p	m	2.1901	1.6575	–	3.4990
Receding Horizon Predictions ($T_h = 2.0$ s, $N = 100$)					
Position error e_p ($T_h = 0.5$ s)	m	0.0072	0.0058	–	0.0231
Position error e_p ($T_h = 1.0$ s)	m	0.0166	0.0135	–	0.0528
Position error e_p ($T_h = 2.0$ s)	m	0.0411	0.0336	–	0.1296
Velocity error e_v ($T_h = 2.0$ s)	m/s	0.0402	0.0338	–	0.1027
Position 95th percentile ($T_h = 2.0$ s)	m	0.0793	–	–	–

Against what reference, and with which parameters, is it validated? The reference is the ground-truth state of the vehicle in the Gazebo simulation, which includes the full six-degree-of-freedom rigid body, the rotor dynamics and drag, and the PX4 inner loops, none of which the model represents. As for the USV, the model is evaluated with the nominal parameters of the simulator and no parameter is fitted to the data, with $m = 2.06$ kg and $g = 9.81$ m/s^2. The thrust is obtained from the normalised command as $T = mg\,u_T/u_{hover}$, as in the controller, and the tether force from (3.46), with $c_d(\varepsilon_0) = 0.2065$ N/m and $\gamma_0 = 28°$ as in Section 4.2.3. The attitude is given to the model in two ways, as measured and as commanded, which separates the error of the translational model from the error of the attitude assumption. The metrics are the RMSE and the coefficient of determination R^2.

What data is used, and what does it cover? The data is a flight of 157 s in which the reference is fed directly to the position controller of PX4 in offboard mode, instead of to the NMPC. As the purpose of the flight is to identify the model and not to evaluate the controller, this gives commands that are independent of the NMPC, and the attitude and thrust generated by the PX4 controller are recorded. The reference was built to excite the motion in all directions. It has two inclined lines flown at 5 and 8 m/s, perpendicular to each other and each climbing from 3 to 10 m, followed by a circle of 5 m radius during which the yaw oscillates by $\pm 69°$ independently of the motion. The vehicle covers 188 m horizontally at up to 8.0 m/s and changes altitude between 2.8 and 9.8 m at up to 2.6 m/s, while the distance to the tether anchor varies from 4.2 to 14.7 m. The commands, shown in Figure 4.15, reach roll and pitch angles of $25.5°$ and $22.6°$, and a thrust between 18.3 and 26.4 N around the weight $mg = 20.2$ N.

Do the assumptions of the model hold in the data? The first assumption is that the attitude is reached as commanded. Figure 4.16 shows that it is reached with a delay: taking the setpoint as the attitude gives an RMSE of $2.27°$ in roll, $2.23°$ in pitch, and $5.88°$ in yaw ($R^2 = 81.5\%$, 75.0%, and 91.4%), with largest deviations of about $20°$ where the command changes abruptly at the transitions between segments of the reference. A first-order lag with a transport delay of about 22 ms explains the residual,

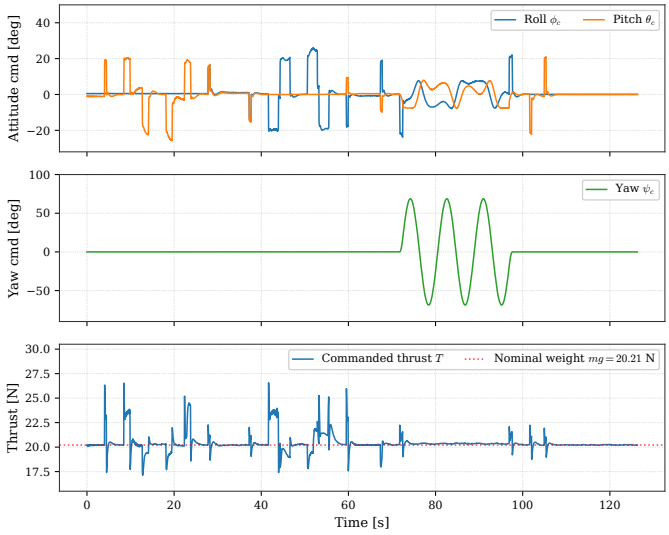


Figure 4.15: UAV attitude and thrust commands during the validation flight.

with time constants of $0.121\,\text{s}$ in roll and pitch and $0.311\,\text{s}$ in yaw, and fits with $R^2 = 98.1\%$, 96.1%, and 97.6%. The second assumption is that the commanded thrust is the thrust produced. The thrust acceleration implied by the measured motion has a mean of $9.905\,\text{m/s}^2$, against $9.901\,\text{m/s}^2$ commanded, which gives a gain of 1.001 between the two, with an RMSE of $0.211\,\text{m/s}^2$ ($R^2 = 73.8\%$). The mapping through the hover thrust is therefore adequate over the whole range of the thrust.

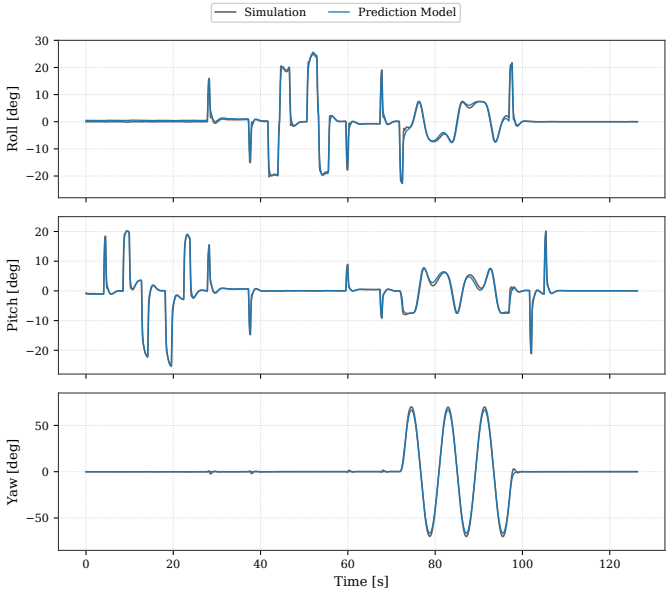


Figure 4.16: UAV attitude tracking. Commanded and measured angles, and the identified first-order lag with delay.

How closely does the model reproduce the reference, in the way it is used? At the level of the acceleration, shown in Figure 4.17, the model driven by the measured attitude reproduces the motion with an RMSE of 0.397, 0.418, and $0.200\,\text{m/s}^2$ along x, y, and z ($R^2 = 83.0\%$, 85.6%, and 76.2%). Driven by the commanded attitude, the horizontal errors become 0.632 and $0.691\,\text{m/s}^2$. In the use that

matters for the controller, the open-loop prediction over the horizon of 1 s, from the measured position and velocity, Figure 4.18 and Table 4.6 show that the position error at the end of the horizon is 0.148 m in RMSE with the measured attitude and 0.203 m with the commanded one, with 95th percentiles of 0.32 and 0.45 m and velocity errors of 0.263 and 0.348 m/s, for displacements of up to 8 m along the trajectory (Figure 4.19).

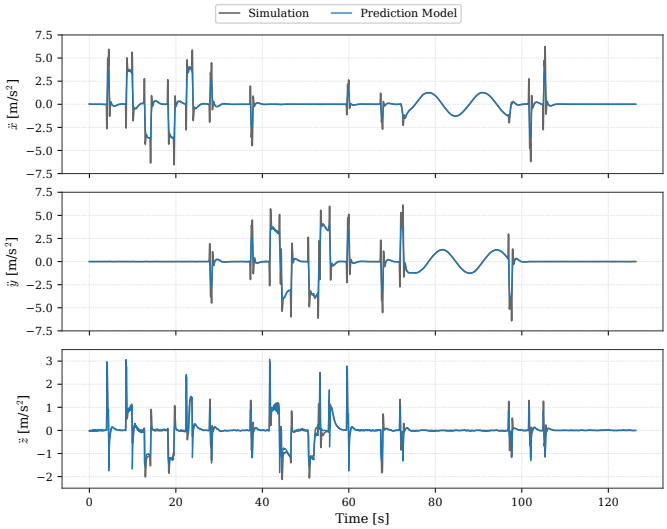
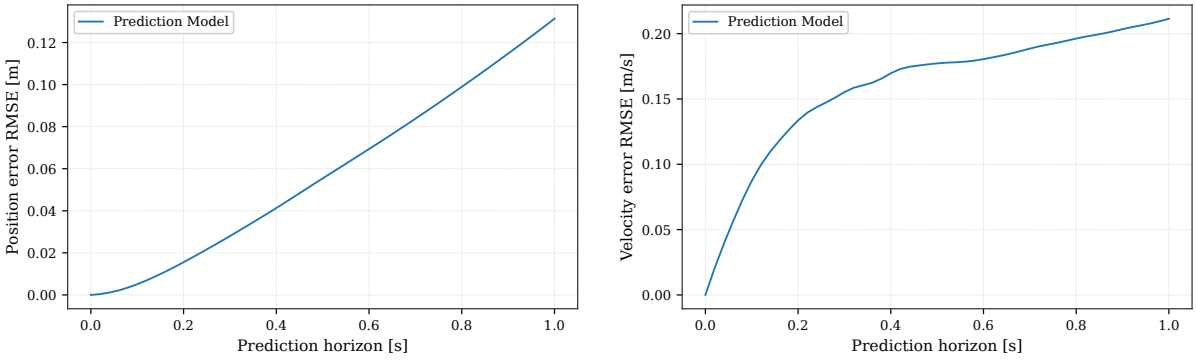


Figure 4.17: UAV inertial acceleration. Ground truth and model driven by the measured attitude.



(a) Position error RMSE.

(b) Velocity error RMSE.

Figure 4.18: UAV open-loop prediction error RMSE as a function of the prediction horizon.

Table 4.6: UAV open-loop prediction error at the end of the 1 s horizon.

Configuration	**Position RMSE (m)**	xy **RMSE (m)**	z **RMSE (m)**	**Position P95 (m)**	**Velocity RMSE (m/s)**
Measured attitude, no tether	0.148	0.132	0.068	0.323	0.263
Measured attitude, tether	0.417	0.245	0.338	0.663	0.822
Commanded attitude, no tether	0.203	0.186	0.081	0.448	0.348
Commanded attitude, tether	0.455	0.296	0.345	0.740	0.884
Commanded attitude, no tether, drag	0.190	0.171	0.081	0.444	0.316

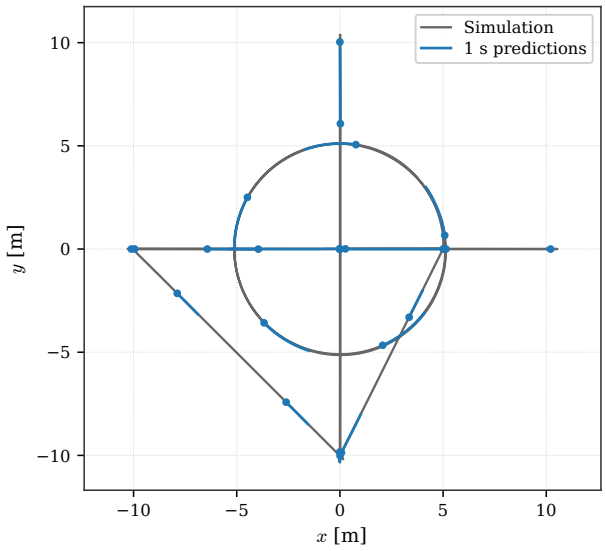


Figure 4.19: One-second open-loop predictions along the UAV flight trajectory.

What explains the remaining error, and is the model adequate for its purpose? Three effects account for it. The largest is the attitude delay, which explains the difference between the commanded and the measured attitude, an additional $0.055\,\mathrm{m}$ of position error after $1\,\mathrm{s}$. The error concentrates at the transitions between segments of the reference, where the acceleration changes abruptly and the position controller commands tilt steps of up to 20° that the vehicle follows with the lag identified above, and it is much smaller in the cruise and in the circle. The second is the aerodynamic drag, which the model does not include: a linear drag $\ddot{\mathbf{p}}_{xy} = -k\,\dot{\mathbf{p}}_{xy}$ with $k = 0.100\,\mathrm{s}^{-1}$ accounts for only 11% of the horizontal residual here and reduces the error with the commanded attitude to $0.190\,\mathrm{m}$. The third is the tether term, which behaves differently. The closed-form force follows the force logged by the coupling node, with a three-dimensional RMSE of $0.424\,\mathrm{N}$ and a mean of $[-0.207,\ 0.120,\ -1.385]^\top\,\mathrm{N}$ against $[-0.134,\ 0.086,\ -1.383]^\top\,\mathrm{N}$, which is about 8% of the weight. However, adding it to the model increases the position error to $0.455\,\mathrm{m}$, because only a small fraction of the logged force appears in the motion of the vehicle: a joint fit of the horizontal residual to the logged force and to the drag gives a gain of 0.11 for the force. The validation therefore supports the model without the tether term, and the effect of the tether on the vehicle in the simulator is evaluated separately in Section 4.2.3. For the controller, the point-mass model with the nominal thrust mapping predicts the position within $0.15\,\mathrm{m}$ with the measured attitude and $0.20\,\mathrm{m}$ with the commanded one, and below $0.45\,\mathrm{m}$ in 95% of the windows, over its horizon, which the feedback of the measured state corrects at every sampling instant, and it is adequate without further identification within the envelope of the recording, up to $8\,\mathrm{m/s}$ and attitudes of 26°.

4.2.3 Tether Model Validation

The control architecture developed in Chapter 3 requires an estimate of the three-dimensional force $\mathbf{F}^{\mathcal{I}}_{\mathrm{tether}}$ exerted by the cable on the UAV to compensate for tether coupling and enforce tension constraints. In simulation, the cable is represented by a high-fidelity lumped-mass model in MoorDyn, which captures multi-node elasticity, bending stiffness, and hydrodynamic drag. However, evaluating such a

high-degree-of-freedom representation or solving the non-linear catenary boundary value problem numerically at every control cycle or across the prediction horizon of an MPC is computationally prohibitive. A closed-form, control-oriented model is therefore essential to provide instantaneous force predictions with negligible computational overhead.

The first-principles formulation derived in Section 3.1.3 computes the tether force directly from its nominal physical specifications under steady winch regulation. With the winch maintaining the slack ratio around $\varepsilon_0 = 0.05$, the linear mass density $\mu = 0.020\,\text{kg/m}$ and gravitational acceleration $g = 9.81\,\text{m/s}^2$ yield a nominal tension-distance coefficient of

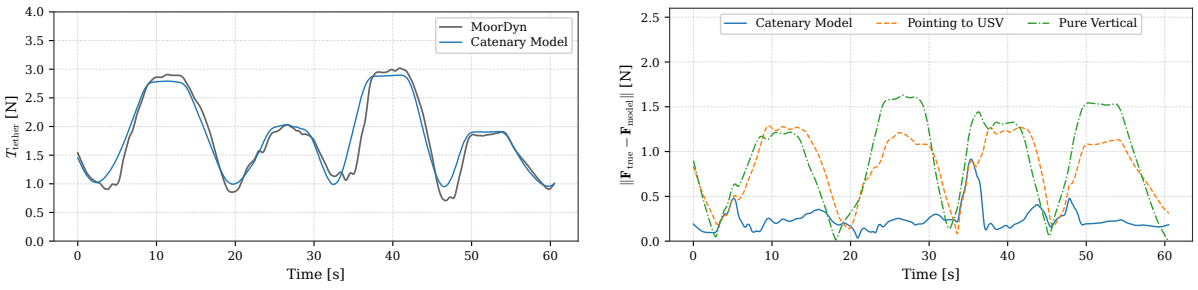
$$c_d(\varepsilon_0) = \frac{\mu\, g}{1 - \varepsilon_0} = \frac{0.020 \times 9.81}{1 - 0.05} \approx 0.2065\,\text{N/m}, \tag{4.6}$$

giving the scalar tension estimate $T_{\text{theo}}(d) = c_d(\varepsilon_0)\, d$, where $d = \|\mathbf{p}_v - \mathbf{p}\|$ is the Euclidean distance between the USV and the UAV. In terms of the unstretched cable length L, this corresponds to $T_{\text{theo}}(L) = \mu\, g\, L = 0.1962\, L$. The three-dimensional direction $\hat{\mathbf{d}}$ is constructed by tilting the line-of-sight chord $\hat{\mathbf{d}}_0$ towards the vertical by the nominal catenary elevation deflection angle $\gamma(\alpha) = \gamma_0 \cos\alpha$, where $\gamma_0 = 28.0°$ is the nominal deflection at $\varepsilon_0 = 0.05$ and α is the chord depression angle. To assess whether this nominal model adequately captures the coupling dynamics, its predictions are evaluated against telemetry from the 30-segment MoorDyn simulator recorded during a $60.5\,\text{s}$ flight trajectory with separation distances ranging from $4.6\,\text{m}$ to $14.0\,\text{m}$. Furthermore, the formulation is benchmarked against two widespread simplifications from the tethered multirotor literature.

1. **Pure Vertical Ballast**, which assumes that the tether hangs directly downward as an equivalent vertical load with $\mathbf{F}_{\text{pure_z}} = [0,\, 0,\, -T_{\text{theo}}]^T$, neglecting horizontal coupling.

2. **Pointing to USV (Straight Chord)**, which assumes a taut cable between the attachment points with $\gamma = 0°$, directing the full tension along the direct line of sight with $\mathbf{F}_{\text{chord}} = T_{\text{theo}}\, \hat{\mathbf{d}}_0$.

Figure 4.20a compares the predicted scalar tension magnitude $T_{\text{theo}}(t)$ with the filtered MoorDyn telemetry over the trajectory. The theoretical linear relation with $c_d = 0.2065\,\text{N/m}$ follows the actual tension closely across the entire operating range, achieving a coefficient of determination of $R^2 = 91.2\%$ and an RMSE of $0.2031\,\text{N}$.



(a) Tether tension magnitude $T_{\text{tether}}(t)$.

(b) Three-dimensional force prediction error norm.

Figure 4.20: Tether tension magnitude and three-dimensional prediction error norm compared against MoorDyn telemetry.

When resolved into Cartesian components in the inertial frame in Figure 4.21, the importance of accounting for the catenary deflection angle $\gamma_0 = 28.0°$ becomes evident. The proposed formulation accurately tracks both the horizontal restoring force F_x with $R^2 = 97.6\%$ and RMSE $= 0.1457$ N, as well as the vertical downward pull F_z with $R^2 = 86.9\%$ and RMSE $= 0.2339$ N. In contrast, the pure vertical simplification fails completely to capture horizontal interaction forces with $R^2 = -0.7\%$, while the straight chord approximation directs excess tension into the horizontal axis and underestimates vertical load because it ignores the natural sag of the cable.

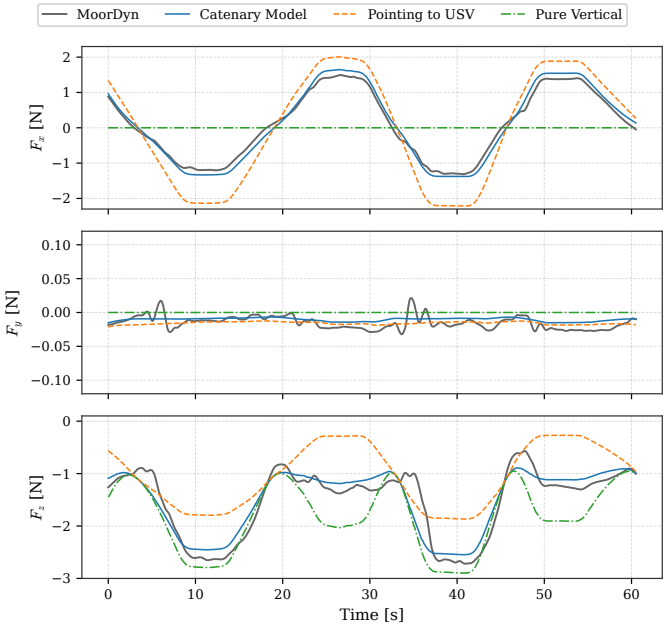


Figure 4.21: Three-dimensional tether force components at the UAV.

The temporal evolution of the three-dimensional prediction error norm $\|\mathbf{F}_{\text{true}} - \mathbf{F}_{\text{model}}\|$, presented in Figure 4.20b, reveals the stability of each approach during dynamic manoeuvres. While the baseline simplifications experience large transient error surges exceeding 1.6 N during vehicle reversals, the proposed catenary model maintains a low and bounded residual below 0.4 N throughout the entire flight profile.

Table 4.7 summarises the quantitative performance across the complete benchmark trajectory. The proposed first-principles model achieves an overall three-dimensional RMSE of 0.2757 N, reducing the spatial force error by more than 70% compared to both the straight chord model with 0.8991 N ($+226.1\%$ error) and the pure vertical model with 1.0332 N ($+274.7\%$ error). These results confirm that the closed-form catenary approximation captures the dominant coupling forces with sufficient fidelity for real-time feedforward compensation, while avoiding the prohibitive computational cost of numerical catenary solvers or multi-segment models.

Table 4.7: Tether force model benchmark.

Model	F_x R^2 (%)	F_x **RMSE** (N)	F_z R^2 (%)	F_z **RMSE** (N)	**3D RMSE** (N)	**Error Increase**
Proposed Catenary	$\mathbf{97.6}$	$\mathbf{0.1457}$	$\mathbf{86.9}$	$\mathbf{0.2339}$	$\mathbf{0.2757}$	**Baseline**
Pointing to USV ($\gamma = 0°$)	57.4	0.6135	-3.8	0.6572	0.8991	$+226.1\%$
Pure Vertical ($F_z = -T$)	-0.7	0.9438	57.6	0.4200	1.0332	$+274.7\%$

4.3 Controller Implementation

4.3.1 Embedded Optimization Framework (CasADi & Acados Pipeline)

4.3.2 USV Controller Implementation

4.3.3 UAV Controller Implementation

4.4 Maneuvers and Missions

4.4.1 Coordinated Maneuver Primitives

4.4.2 Autonomous Missions and Coordination Logic

4.5 Software Architecture and Dashboard

4.5.1 Operator Web Dashboard and Mission Management

4.5.2 Integrated Software Architecture and System Dataflow

Chapter 5

Conclusions

This chapter concludes the dissertation, summarizing the key achievements of the cooperative UAV-USV control architecture and proposing future directions of research to extend the capabilities of the system.

5.1 Summary of Achievements

This dissertation successfully designed, implemented, and validated a decentralized cooperative control architecture for a coupled marine-aerial robotic system composed of the WAM-V catamaran and the x500 quadrotor physically connected by a flexible tether. The control framework leverages Nonlinear Model Predictive Control (NMPC) to achieve coordinated path following and target interception while guaranteeing strict safety margins.

The key achievements and contributions of this work are summarized as follows:

- **Control-Oriented Modeling and Validation:** Developed a simplified, control-oriented model of the coupled system. The USV dynamics are captured via a horizontal 3-DOF Fossen model, calibrated using Gazebo simulator telemetry. The UAV dynamics incorporate a first-order attitude lag model ($\tau = 0.15\,\text{s}$) representing the closed-loop response of the autopilot, which was successfully identified and validated against PX4 flight telemetry.
- **Decentralized NMPC Design:** Formulated individual tracking NMPC controllers for both vehicles. The USV controller operates at $10\,\text{Hz}$ to calculate virtual force demands, while the UAV controller runs at $50\,\text{Hz}$ to compute collective thrust and attitude rates. Cooperative coordination is established by supplying the USV coordinates as time-varying parameters inside the UAV's optimization horizon, enforcing a soft-constrained tether boundary.
- **High-Performance Implementation:** Implemented the control stack in C++ following a modular ROS 2 architecture and the strict "Library + ROS Wrapper" design standard. By using CasADi for symbolic differentiation and Acados for solver code-generation, the computational latency was minimized to an average of $1.5\,\text{ms}$ for the UAV and $0.8\,\text{ms}$ for the USV, proving real-time feasibility.

- **Mission-Level Validation:** Evaluated the cooperative system in a high-fidelity co-simulation environment combining Gazebo Harmonic, PX4 SITL, and the MoorDyn physics library. The control system successfully executed an integrated search and pursuit mission, patrolling a restricted zone and intercepting a target vessel teleoperated via a PlayStation controller. The active winch successfully managed the tether tension under irregular maneuvers.

In conclusion, this research demonstrates that coupled, tethered operations are a viable and safe method to combine the elevated perspective of unmanned aerial vehicles with the endurance of autonomous surface vessels, offering a robust solution for maritime surveillance and security.

5.2 Future Work

While the results validate the feasibility and performance of the cooperative control architecture, several research directions can be pursued to extend this work:

- **Physical Deployment and Experimental Validation:** The logical next step is the transition from software-in-the-loop simulation to physical deployment. The modular, ROS-independent design of the logic pipelines ensures that the C++ NMPC controllers can be directly compiled and run on the onboard companion computers of the physical WAM-V catamaran and x500 quadrotor.
- **Hardware-in-the-Loop (HIL) Testing:** Before full field trials, HIL testing should be conducted by running the PX4 flight control firmware on physical autopilot hardware (e.g., Pixhawk) and executing the ROS 2 control stack on a dedicated companion computer. This will validate the controller under real processor constraints, sensor noise, and communication latency.
- **Disturbance Observer Design:** Integrating estimator algorithms (such as Extended Kalman Filters or disturbance observers) would allow the NMPC controllers to estimate external wind and wave forces in real-time. Feeding these estimated disturbances back into the parameters of the optimization solvers would improve trajectory tracking accuracy under severe environmental conditions.
- **Resilient Decentralized Communication:** Investigate the robustness of the decentralized coordinator under communication failures, packet loss, or latency in the USV-to-UAV data link, developing fail-safe behaviors where the vehicles fall back to local safety controllers.
- **Tether Power Delivery and Energy Harvesting:** Explore the physical optimization of power transfer through the tether to supply the UAV from the catamaran's main batteries, extending flight times indefinitely. Additionally, energy regeneration strategies could be investigated, utilizing the winch motor to harvest energy from the pulling forces exerted by the UAV during high-tension maneuvers.

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